

Searching for the digital dividend: Internet diffusion and productivity growth in 149 economies, 1996–2025

Abstract

We examine conditional associations between lagged internet use and labour-productivity growth in 149 economies over 1996–2025. Within countries, years with a higher internet-user share were not years of faster growth. A linear fixed-effects estimate of -0.284 percentage points per ten-percentage-point higher share is robust to country-specific trends and convergence controls. A cross-fitted machine-learning estimate of -0.264 is less precise, and its 95% interval includes zero under two-way clustering. The association is concentrated in 1996–2011. To study heterogeneity we use Panel-DOSE, a debiased machine-learning spline estimator for fixed-effects panels. The association does not vary robustly with schooling, initial productivity or time since internet take-off, although the data cannot rule out moderate gradients. A gradient along initial productivity appears in both Penn World Table releases but is much weaker in World Bank data for the same country-years; about half of this difference comes from the growth series, and it is imprecisely estimated. These observational results do not identify the causal return to connectivity investment.

Keywords: Internet diffusion, Labor productivity, Double machine learning, Heterogeneous effects, Panel data, Data revisions

JEL: C14, C23, C55, O33, O47

1. Introduction

The expectation that digital technologies deliver a “digital dividend” in the form of faster productivity growth is central to development strategies and to much of the digital-economy literature (World Bank, 2016; Goldfarb and Tucker, 2019). The empirical record is more mixed than this expectation suggests. Early macro studies of telecommunications and broad-

band report large growth effects (Röller and Waverman, 2001; Koutroumpis, 2009; Czernich et al., 2011), and growth accounting attributes much of the late-1990s US productivity revival to information technology (Jorgenson et al., 2008; Jorgenson and Vu, 2016). Firm-level studies find large returns mainly where technology is combined with organisational change and skilled labour (Brynjolfsson and Hitt, 2000; Bresnahan et al., 2002; Bloom et al., 2012), while cross-country evidence for developing and emerging economies is weaker (Niebel, 2018; Vu et al., 2020; Cardona et al., 2013). Recent contributions to this journal likewise emphasise that the consequences of digitalisation depend on the structure of the economy and on how digital inputs enter production (Banga et al., 2026; Chen and Li, 2026).

A common thread is heterogeneity. The returns to a new technology are expected to depend on the capacity to absorb it, proxied by human capital (Nelson and Phelps, 1966; Benhabib and Spiegel, 1994; Cohen and Levinthal, 1990); on distance to the technological frontier (Acemoglu et al., 2006); and on how long the technology has been in use, because general-purpose technologies require complementary intangible investment that depresses measured productivity before raising it (Brynjolfsson et al., 2021). Each argument predicts a specific shape of the function linking the effect of adoption to a characteristic of the adopting economy. Testing these predictions requires an estimator that recovers that function without imposing a linear interaction and without a linear control function, because the determinants of productivity growth and of adoption—income, demography, openness, urbanisation—interact nonlinearly.

This paper examines whether, over three decades of internet diffusion, a productivity dividend is visible in aggregate data, and whether the conditional association between adoption and growth varies in the ways these theories predict. We do not claim to identify the causal effect of adoption: the design is observational, and the estimates are associations conditional on controls and fixed effects. We combine World Bank productivity data and International Telecommunication Union (ITU) adoption data for 149 economies over 1996–2025 with schooling data from the United Nations Development Programme (UNDP), and

we compare the results with two releases of the Penn World Table (PWT) (Feenstra et al., 2015; Groningen Growth and Development Centre, 2025). To estimate how the association varies with a moderator we use Panel-DOSE (Panel Debiased Orthogonal Sieve Estimator), which approximates the varying coefficient by a penalised spline and estimates it from Neyman-orthogonal moments with machine-learning nuisance functions, Mundlak features, cross-fitting by country and a two-way within transformation.

Three findings stand out. First, within countries, a higher lagged internet-user share is not associated with faster labour-productivity growth. A linear fixed-effects regression gives -0.284 percentage points (pp) per ten-percentage-point higher share, and this estimate is stable when we add country-specific trends, lagged productivity or development-specific year effects. The flexible estimate is -0.264 pp, but its interval includes zero with two-way clustering and for about a third of the random seeds of the sample splits, so we do not claim a significant negative association. The association is concentrated in 1996–2011, is weaker without the transition economies, and is shared with the co-diffusing mobile and broadband series. By contrast, a cross-country long difference shows that economies with larger increases in internet use grew faster between 2000 and 2024, a comparison that fixed effects remove and that is consistent with faster-growing economies adopting faster. Second, we find no robust variation of the within-country association with schooling, with initial productivity or with the time elapsed since internet take-off. None of the three primary tests rejects, but the minimum detectable gradients are moderate, so the data cannot rule out complementarities of that size. Third, estimates of heterogeneity depend on the productivity data. With the same estimator, the gradient along initial productivity is large in both releases of the Penn World Table and much weaker in World Bank data. On the country-years common to both sources, about half of the difference comes from the growth series and a third from the control set, but a paired bootstrap cannot exclude that the sources agree. The revision from PWT 10.0 to PWT 11.0 does not change the gradient detectably. This complements the evidence of Johnson et al. (2013) that the choice of data changes growth results.

The paper also makes a methodological contribution, which we state modestly. Panel-DOSE combines known components: orthogonal series estimation of causal functions (Semenova and Chernozhukov, 2021), the correlated-random-effects and within-group variants of panel double/debiased machine learning (DML) (Clarke and Polsell, 2026), cross-fitting by clusters, and O’Sullivan penalised splines (O’Sullivan, 1986; Wand and Ormerod, 2008). We show that the combination is Neyman-orthogonal, that the within step makes the moment invariant to additive country or year errors in the outcome nuisance, that when the moderator is a control only two nuisance functions are needed whatever the flexibility of the sieve, and that the tests of constancy and linearity remain valid for a fixed penalty. Relative to the panel heterogeneous-effect estimator of Semenova et al. (2023), which targets a finite-dimensional projection of the conditional effect under weak time dependence, Panel-DOSE targets a nonparametric function of a single moderator with a continuous treatment under two-way fixed effects and cluster sampling; relative to Klosin and Vilgalys (2022) and Colangelo and Lee (2026), who estimate average partial effects or dose-response functions, it targets effect heterogeneity. Monte Carlo experiments, including a plasmode design built on the empirical panel for each moderator (Franklin et al., 2014), show where the estimator helps and where it does not. In the stylised designs it has the lowest integrated error among the feasible estimators we compare when the effect varies and the confounding is strongly nonlinear, and most of the gain comes from the Mundlak features. A two-way fixed-effects (TWFE) regression with spline controls does as well when the confounding is only mildly nonlinear. Linear interaction models produce spurious gradients in the plasmode design built on our data, and the penalised bands under-cover curved effect functions, so we use them descriptively.

Section 2 reviews the literature and states the predictions we test. Section 3 presents the estimator. Section 4 reports the simulations and Section 5 the application. Section 6 discusses interpretation, policy and limitations, and Section 7 concludes. The supplementary material contains proofs (Appendix A), data and implementation details (Appendix B) and

additional results (Appendix C).

2. Related literature and predictions

2.1. ICT, broadband and aggregate growth

Röller and Waverman (2001) estimate a structural model of telecommunications investment and output for OECD countries and find large growth effects of network roll-out; Koutroumpis (2009) obtains similar results for broadband. Czernich et al. (2011) instrument broadband penetration with pre-existing voice and cable networks and find that a ten-point increase raised annual per-capita growth by 0.9–1.5 pp in OECD countries. Growth accounting documents large contributions of ICT capital in advanced economies and, increasingly, in emerging economies (Jorgenson et al., 2008; Jorgenson and Vu, 2016). Surveys of the ICT–productivity literature conclude that estimated returns are positive on average but heterogeneous and sensitive to the level of aggregation and to method (Cardona et al., 2013; Vu et al., 2020). Niebel (2018) finds no larger returns to ICT capital in developing than in developed economies. For the developing world, Hjort and Poulsen (2019) show that the arrival of submarine cables raised employment in Africa, and Hjort and Tian (2025) review a literature that finds positive effects of connectivity on employment and firm outcomes but little direct evidence on aggregate productivity.

2.2. Complementarities, diffusion and capabilities

At the firm and worker level, the productivity gains from digital technologies depend on skills and organisation. Broadband raised the productivity of skilled workers and lowered that of unskilled workers in Norway (Akerman et al., 2015); US multinationals obtained higher returns to IT than comparable European firms because of their management practices (Bloom et al., 2012). OECD work finds that digital adoption and its productivity pay-off are concentrated in firms and industries with the capabilities to use it (Gal et al., 2019; Nicoletti et al., 2020). At the country level, Comin and Hobijn (2010) and Comin and Mestieri (2018) show that adoption lags have fallen but the intensity of use still differs

widely across countries, so that access and use diverge. Recent contributions to this journal study how digitalisation relates to structural change across countries (Banga et al., 2026) and how data and digital inputs can be represented in aggregate production functions (Chen and Li, 2026); the productivity channel we study complements the structural-change channel emphasised by Banga et al. (2026). Machine-learning methods are also increasingly used in this journal to predict firm outcomes (Xu et al., 2026); we use them instead to control flexibly for confounders when estimating a heterogeneous association.

2.3. Measurement and data revisions

Cross-country growth regressions are sensitive to the release of the data. Johnson et al. (2013) show that growth rates in successive PWT releases differ substantially for the same countries and years, and that the revisions change the results of growth regressions; Ponomareva and Katayama (2010) show the same for the growth–volatility relationship. Mis-measurement of digital output has also been proposed as an explanation for the post-2005 productivity slowdown, although it accounts for only a small part of it (Byrne et al., 2016; Syverson, 2017). Our vintage comparison applies this insight to heterogeneous associations of internet diffusion and attributes the change between releases to the revised outcome, moderator and controls.

2.4. Machine learning for heterogeneous and continuous effects

Double/debiased machine learning (Chernozhukov et al., 2018) generalises the partialling-out estimator of Robinson (1988). Clarke and Polselli (2026) extend it to static panels with fixed effects through correlated-random-effects, within-group and first-difference implementations of a constant coefficient. Semenova et al. (2023) estimate heterogeneous effects in high-dimensional dynamic panels under weak dependence, targeting the best linear predictor of the conditional effect in a set of covariates. Semenova and Chernozhukov (2021) project orthogonal signals on series bases to estimate causal functions. For continuous treatments, Colangelo and Lee (2026) estimate dose-response functions and partial effects with

kernel-based DML, and Klosin and Vilgalys (2022) estimate average partial derivatives in panels with unit effects. Causal forests and the R-learner estimate conditional effects in cross-sectional data (Wager and Athey, 2018; Athey et al., 2019; Nie and Wager, 2021). The TWFE literature shows that linear fixed-effects regressions with heterogeneous or continuous treatments estimate weighted averages whose weights can be negative or hard to interpret (de Chaisemartin and D’Haultfœuille, 2020; Callaway et al., 2024). Panel-DOSE targets a different estimand from these methods: the function $\theta(z)$ describing how the association of a continuous treatment with the outcome varies with a scalar moderator, under two-way fixed effects and dependence within countries. Its components are known; what the paper adds is their combination for this estimand, the associated properties, and evidence on when the combination helps.

2.5. Predictions

We translate the theories into predictions on the shape of $\theta(\cdot)$ in model (1) below. Absorptive capacity (Nelson and Phelps, 1966; Benhabib and Spiegel, 1994; Akerman et al., 2015) predicts that θ increases with schooling. Leapfrogging arguments, in which the internet lowers information and transaction costs most where they are highest (Hjort and Poulsen, 2019), predict that θ decreases with initial productivity; frontier-based arguments (Acemoglu et al., 2006) allow either sign. The productivity J-curve (Brynjolfsson et al., 2021) predicts that θ is negative in the first years after a country’s adoption takes off and rises with time since take-off. Network effects predict a marginal effect of adoption $f'(d)$ that rises with the level of adoption; saturation predicts that it falls. Finally, if the gains accumulate slowly, cumulative associations over several years should exceed the one-year association. Because our design is observational, we test these predictions on the shape of the conditional association, under the assumptions of Section 3; a failure to find a predicted shape is informative only up to the precision we report.

3. Methodology

In words, the estimator proceeds in four steps. It first predicts productivity growth and internet adoption from each country’s characteristics and their country averages, using a machine-learning model trained on other countries. It then keeps the prediction errors—the parts of growth and adoption that the characteristics do not explain—and removes country and year averages from them. Finally, it fits a smooth curve that describes how the unexplained part of growth co-moves with the unexplained part of adoption for different values of a moderator such as schooling. The height of this curve at a given value of the moderator, $\theta(z)$, is the difference in annual growth, in percentage points, associated with a unit (100 percentage points) higher lagged internet-user share; one tenth of it is the difference associated with ten more users per hundred people. Readers interested mainly in the results can skip to Section 5.

3.1. Model and target parameters

We observe a panel $\{(Y_{it}, D_{it}, X_{it}) : i = 1, \dots, G; t = 1, \dots, T_i\}$ of G countries, where Y_{it} is an outcome (productivity growth), $D_{it} \in \mathbb{R}$ is a continuous treatment (the lagged internet-user share) and $X_{it} \in \mathbb{R}^p$ collects controls that include the moderator Z_{it} . The varying-coefficient model is

$$Y_{it} = \theta(Z_{it}) D_{it} + g(X_{it}) + \alpha_i + \lambda_t + \varepsilon_{it}, \quad (1)$$

in which $g(\cdot)$ is an unknown control function and (α_i, λ_t) are country and year effects. Under the assumptions below, $\theta(z)$ is the difference in Y associated with a unit difference in D for an economy with moderator value z , holding the controls and fixed effects constant. We call it an effect for brevity when discussing the estimator, but in the application it is a conditional association (Section 6). The dose-response variant replaces $\theta(Z_{it})D_{it}$ by $f(D_{it})$ and targets $f'(d)$.

We report four summaries of $\theta(\cdot)$. The *headline average* is the constant-effect coefficient $\bar{\theta}_w$ obtained when the sieve has a single basis function, i.e. the fixed-effects DML (FE-DML) estimator of Clarke and Polsell (2026). It equals a weighted average of $\theta(Z_{it})$ with weights proportional to the residual variance of D_{it} given the controls. The *sample-average effect* is $\bar{\theta} = \mathbb{E}[\theta(Z_{it})]$, which weights country-years equally. The two coincide when θ is constant and differ otherwise; we report both and treat $\bar{\theta}_w$ as the headline because it does not depend on the sieve. The *group-average effects* $\bar{\theta}_A = \mathbb{E}[\theta(Z_{it}) \mid Z_{it} \in A]$ use the terciles of Z , and the *best linear projection (BLP) slope* is $\gamma = \text{Cov}(\theta(Z_{it}), Z_{it})/\text{Var}(Z_{it})$, computed over the estimation sample.

3.2. Identification

Write $F_{it} = (X_{it}, \bar{X}_i, \bar{D}_i, \bar{Z}_i, t)$ for the controls augmented by their country averages and a time index.

Assumption 1 (Conditional strict exogeneity). $\mathbb{E}[\varepsilon_{it} \mid D_{i1}, \dots, D_{iT_i}, X_{i1}, \dots, X_{iT_i}, \alpha_i] = 0$ for all t .

Assumption 2 (Correlated random effects in the nuisance step). $\mathbb{E}[\alpha_i \mid F_{it}] = h(\bar{X}_i, \bar{D}_i, \bar{Z}_i)$ for some function h that does not depend on t .

Assumption 3 (Nuisance sufficiency). $\mathbb{E}[D_{it} \mid F_{i1}, \dots, F_{iT_i}] = \mathbb{E}[D_{it} \mid F_{it}]$ for all t .

Assumption 1 is the strict-exogeneity condition of static fixed-effects models. It fails mechanically if the controls contain lagged levels or growth of the outcome variable: with $Y_{it} = \Delta \ln \text{LP}_{it}$, the control $\ln \text{LP}_{i,t}$ that enters at $t + 1$ contains Y_{it} , and so does its country mean. Our baseline therefore excludes lagged productivity levels and lagged growth, uses initial (pre-sample) productivity instead, and forms Mundlak averages only of controls that are not functions of the outcome. Assumption 2 is a nonparametric version of the Mundlak device (Mundlak, 1978; Wooldridge, 2010): the conditional mean of the country effect given the augmented controls may be any function of the country averages. The part of α_i it leaves

unexplained, $a_i = \alpha_i - h(\cdot)$, is removed by the within transformation. Assumption 3 requires that the augmented controls of period t summarise the information in the whole control path that is relevant for predicting adoption; the Mundlak averages are included to carry that country-level information. Finally, D must vary after conditioning on F and removing country and year means; we report this residual variation in Section 5.

Let $b(z) = (b_1(z), \dots, b_K(z))'$ be a cubic B-spline basis with interior knots at quantiles of Z , write $\theta(z) = b(z)'\beta + r_K(z)$, define the sieve regressors $\Phi_{it} = D_{it}b(Z_{it})$, and the nuisance functions $\ell(F) = \mathbb{E}[Y \mid F]$ and $m(F) = \mathbb{E}[\Phi \mid F]$. Let $\mathcal{M}$ be the two-way within operator (for unbalanced panels, computed by alternating projections). The population moment is

$$\psi(\beta; \ell, m) = \mathbb{E}\left[\sum_t \mathcal{M}(\Phi - m(F))_{it} \left\{ \mathcal{M}(Y - \ell(F))_{it} - \mathcal{M}(\Phi - m(F))'_{it}\beta \right\}\right] = 0. \quad (2)$$

Under Assumptions 1–3 and $r_K = 0$, $Y - \ell(F) - (\Phi - m(F))'\beta = a_i + \varepsilon_{it}$, because λ_t is a function of the time index in F . The term a_i is removed by $\mathcal{M}$ and ε_{it} is mean-independent of the whole path of F , so the true β solves (2).

3.3. Properties

Proposition 1 (Orthogonality and invariance). *Let Assumptions 1–3 hold. (i) Neyman orthogonality: for any directions (δ_ℓ, δ_m) , $\partial_r \psi(\beta_0; \ell_0 + r\delta_\ell, m_0 + r\delta_m)|_{r=0} = 0$. (ii) Invariance for the outcome nuisance: for any country-specific and year-specific constants c_i, c_t , $\psi(\beta; \ell + c_i + c_t, m) = \psi(\beta; \ell, m)$ for all β . The same holds for additive errors $d_i + d_t \in \mathbb{R}^K$ in m . When $m = b(Z)\mu$ (Proposition 2), an error c_i in μ enters the regressors as $\mathcal{M}\{b(Z_{it})c_i\}$; this term is zero if $b(Z_{it})$ is constant within each country (in particular if $K = 1$ or Z is time-invariant), and in general it is not zero otherwise.*

Part (i) makes the estimator insensitive to first-order errors in the machine-learning fits. Part (ii) says that any error in the outcome learner that takes the form of a country or year shift—a natural failure mode when trees approximate fixed effects through country averages—has no effect on the estimate. For the treatment nuisance the invariance is guaran-

teed for the constant-effect estimator and for time-invariant moderators; with a time-varying moderator, country-level errors in $\hat{\mu}$ can enter, although only at second order by part (i). A numerical check on the empirical sample confirms both statements (Appendix B).

Proposition 2 (Structure-aware residualisation). *If Z_{it} is an element of F_{it} , then $m_k(F) = b_k(Z) \mu(F)$ with $\mu(F) = \mathbb{E}[D \mid F]$. Hence the orthogonalised sieve regressors are $b(Z_{it})\{D_{it} - \mu(F_{it})\}$, and only ℓ and μ need to be learned, whatever the sieve dimension K .*

Proposition 2 follows from $\mathbb{E}[D b_k(Z) \mid F] = b_k(Z) \mathbb{E}[D \mid F]$. It reduces the number of learners from $K + 1$ to two, avoids fitting K separate regressions for highly collinear spline regressors, and allows the sieve (number of knots, penalty, or a linear basis) to be changed after the nuisance step without refitting. It also implies that the constant-effect FE-DML estimate is obtained from the same residuals. In the dose-response model, Φ is a basis in D and the K conditional means are learned separately.

3.4. The Panel-DOSE algorithm

The final stage minimises a penalised least-squares criterion in the orthogonalised variables,

$$\hat{\beta}_\lambda = \arg \min_{\beta} \sum_{i,t} (\tilde{Y}_{it} - \tilde{\Phi}'_{it} \beta)^2 + \lambda \kappa \beta' \Omega \beta, \quad \Omega = \int b''(z) b''(z)' dz, \quad (3)$$

where $\tilde{Y} = \mathcal{M}(Y - \hat{\ell})$, $\tilde{\Phi} = \mathcal{M}(\Phi - \hat{m})$, Ω is the O'Sullivan penalty (O'Sullivan, 1986; Wand and Ormerod, 2008) and $\kappa = \text{tr}(\tilde{\Phi}' \tilde{\Phi}) / \text{tr}(\Omega)$ makes λ scale-free. Because $\beta' \Omega \beta$ is the integrated squared second derivative of $b(z)' \beta$, its null space consists exactly of the linear functions, whatever the knot placement, so a large λ shrinks θ towards a linear function.¹ We choose λ by ten-fold cross-validation in which the folds are formed by countries, over $\lambda \in \{0\} \cup \{10^{-6}, \dots, 10^4\}$, and report the selected λ and the effective degrees of freedom $\text{edf} = \text{tr}\{(\tilde{\Phi}' \tilde{\Phi} + \lambda \kappa \Omega)^{-1} \tilde{\Phi}' \tilde{\Phi}\}$. At the upper end of the grid the fit is effectively linear

¹The second-difference penalty of Eilers and Marx (1996) has this property only for equally spaced knots; with knots at quantiles of a skewed moderator its null space contains nonlinear functions.

(edf ≈ 2). To avoid reporting a median over splits that selected very different amounts of smoothing, we choose one λ for all splits, the minimiser of the cross-validation criterion summed over the S splits. Algorithm 1 in the supplementary material summarises the procedure.

Cross-fitting by countries rather than by observations is essential in a panel. The Mundlak means are computed within each country, so with folds formed by countries no feature of a training country contains information about a test country; using the country mean of D , which includes future adoption, is justified by strict exogeneity (Assumption 1). With observation-level folds, the learner is trained on other years of the same country and partly reproduces its own country effect. Repeating the split S times and aggregating by the median, functional by functional, with the variance adjustment of Chernozhukov et al. (2018), reduces the sensitivity of the results to a particular partition. Any linear transformation of the functionals—centring or detrending a grid, taking a contrast—is applied to each split’s coefficients before the median is taken, so that every reported number is aggregated in the same way. The element-wise median of the covariance matrices is projected on the positive semi-definite cone; its diagonal equals the median-aggregated variances. Because the median of a difference need not equal the difference of the medians, a reported contrast can differ slightly from the difference of the reported group averages.

3.5. Inference

Let $A_\lambda = \tilde{\Phi}'\tilde{\Phi} + \lambda\kappa\Omega$ and $\hat{u}_{it} = \tilde{Y}_{it} - \tilde{\Phi}'_{it}\hat{\beta}$. The cluster-robust covariance estimator is

$$\hat{V} = c_n A_\lambda^{-1} \left(\sum_{i=1}^G \hat{s}_i \hat{s}_i' \right) A_\lambda^{-1}, \quad \hat{s}_i = \sum_t \tilde{\Phi}_{it} \hat{u}_{it}, \quad (4)$$

with $c_n = \frac{G}{G-1} \frac{n-1}{n-K}$ (Cameron and Miller, 2015). Standard errors of linear functionals $w'\beta$ follow directly. Simultaneous bands over a grid use the sup- t critical value from the correlation matrix of the estimated function (Montiel Olea and Plagborg-Møller, 2019). We test whether θ is constant, and whether it is linear, with sup- t statistics on a 25-point grid of

moderator quantiles. The statistic is the maximum absolute t -ratio of the centred (or linearly detrended) function, and its null distribution is simulated from the corresponding, possibly rank-deficient, covariance matrix, so no matrix inversion is required.² For each moderator, the constancy test is the primary test of heterogeneity, and we adjust the three primary tests (one per moderator) by the method of Holm (1979). We designate these tests as primary in this paper; there is no registered pre-analysis plan. Tercile contrasts and BLP slopes are descriptive summaries, and robustness checks are not treated as additional hypotheses. To compare two estimates computed on the same countries, such as those from the two PWT releases, we use a paired country bootstrap that re-estimates the nuisance functions, the penalty and the tercile groups of both estimates in every draw. We also report two-way (country and year) clustered standard errors for the baseline, $c_G V_{\text{country}} + c_T V_{\text{year}} - c_n V_{\text{obs}}$ with $c_G = G/(G - 1)$, $c_T = T/(T - 1)$ and $c_n = n/(n - K)$ (Cameron et al., 2011), replaced by the country-clustered variance if the combination is negative. With about 30 years, the year dimension provides few clusters, so these standard errors are themselves imprecise.

Assumption 4 (Regularity). *(a) Countries are independent, $G \rightarrow \infty$, and $T_i \leq \bar{T} < \infty$. (b) The cross-fitted nuisance estimators satisfy $\|\hat{\mu} - \mu\|_2 (\|\hat{\mu} - \mu\|_2 + \|Q(\hat{\ell} - \ell)\|_2) = o_p(G^{-1/2})$ and $\|\hat{\ell} - \ell\|_2 = o_p(1)$, where Q removes country means: by Proposition 1(ii), only the within-country error of $\hat{\ell}$ matters, whereas the full error of $\hat{\mu}$ enters. (c) $J = \mathbb{E}[\sum_t \tilde{\Phi}_{it}^0 \tilde{\Phi}_{it}^{0'}]$ is positive definite and fourth moments are bounded. (d) K is fixed and $\lambda_G \sqrt{G} \rightarrow 0$.*

Proposition 3 (Asymptotic normality). *Under Assumptions 1–4, Assumption A.1 in Appendix A, and the conditions of Proposition 2, $\sqrt{G}(\hat{\beta} - \beta_K) \xrightarrow{d} N(0, J^{-1} \Omega_s J^{-1})$, where β_K is the sieve pseudo-true value and $\Omega_s = \mathbb{E}[s_i s_i']$ with $s_i = \sum_t \tilde{\Phi}_{it}^0 u_{it}$, and $G\hat{V} \xrightarrow{p} J^{-1} \Omega_s J^{-1}$.*

Proposition 3 requires a vanishing penalty. The cross-validated penalty of the application does not vanish: κ grows with the number of observations, so that a fixed λ corresponds to

²Wald tests on differences of spline coefficients would require inverting a near-singular covariance matrix; the sup- t statistics avoid this.

a penalty of order G in (3). The next result shows that the shape tests remain valid for any fixed penalty, because constant and linear functions lie in the null space of the O’Sullivan penalty.

Proposition 4 (Shape tests under a non-vanishing penalty). *Let the conditions of Proposition 3 hold except that $\lambda \geq 0$ is fixed, and let $\bar{\kappa} = \text{plim } \kappa/G$ and $J_\lambda = J + \lambda \bar{\kappa} \Omega$. Then $\sqrt{G}(\hat{\beta}_\lambda - \tilde{\beta}_\lambda) \xrightarrow{d} N(0, J_\lambda^{-1} \Omega_s J_\lambda^{-1})$, where $\tilde{\beta}_\lambda = (\tilde{\Phi}' \tilde{\Phi} + \lambda \kappa \Omega)^{-1} \tilde{\Phi}' \tilde{\Phi} \beta_K$ is the penalised projection of the sieve coefficients. If θ is linear in the winsorised moderator, then $\Omega \beta_K = 0$, so $\tilde{\beta}_\lambda = \beta_K$: the penalised estimator has no smoothing bias, $G\hat{V} \xrightarrow{p} J_\lambda^{-1} \Omega_s J_\lambda^{-1}$, and the sup- t linearity test has asymptotic size equal to its nominal level. The same holds for the constancy test when θ is constant, and, in the dose-response model, for the test that f' is constant (linear f).*

Remark 1. *Propositions 3 and 4 are fixed- K results. Proposition 3 gives inference on the pseudo-true function $b(z)'\beta_K$ and on its functionals—group averages, contrasts and the BLP slope—not on $\theta(z)$ itself. With a non-vanishing penalty, Proposition 4 centres the estimator at the penalised projection $\tilde{\beta}_\lambda$, which depends on the sample design matrix; off the null, intervals for penalised functionals are therefore conditional on the design and omit the variability of $\tilde{\beta}_\lambda$ around its limit, and they do not cover a curved θ at the nominal rate. Two further qualifications apply. First, the moderator is winsorised at its 0.5th and 99.5th percentiles before the basis is formed, so the null hypotheses concern θ on that range (a linear θ is linear in the winsorised moderator). Second, the theory treats λ as fixed, whereas the application selects it by cross-validation. We therefore calibrate the tests by simulation (Section 4.3), report their simulated size alongside the p-values, and read the bands as descriptions of the penalised fit rather than as confidence sets for θ . Inference on $\theta(z)$ with $K \rightarrow \infty$ would require the conditions of Belloni et al. (2015) and Semenova and Chernozhukov (2021), which we do not verify for panels.*

4. Monte Carlo evidence

4.1. Stylised design

The first design mimics a macro panel of $G = 100$ countries observed for $T = 24$ years. Each country has a fixed effect $\alpha_i \sim N(0, 1)$ and there is a random-walk year effect. Five controls follow $X_{it} = 0.6X_{i,t-1} + 0.4(\eta_{it} + 0.5\alpha_i)$ with $\eta_{it} \sim N(0, I_5)$. The moderator $Z_{it} \in [0, 1]$ trends slowly around a country-specific level that is correlated with the country effect through $\mathbb{I}\{\alpha_i > 0\}$. Adoption follows a logistic diffusion index,

$$D_{it} = \Lambda\left(-3 + 0.22t + 0.6\alpha_i + 0.8\sin(1.5X_{1it}) + 0.6X_{2it}\mathbb{I}\{X_{3it} > 0\} + 1.5Z_{it} + v_{it}\right),$$

where Λ is the logistic function and $v_{it} \sim N(0, 1)$, and the outcome is $Y_{it} = \theta(Z_{it})D_{it} + g(X_{it}, Z_{it}) + \alpha_i + \lambda_t + \varepsilon_{it}$ with $\varepsilon_{it} \sim N(0, 0.25)$ and

$$g = \rho\{1.5\sin(1.5X_{1it}) + X_{2it}\mathbb{I}\{X_{3it} > 0\} + 0.5|X_{4it}| + 3.2(Z_{it} - 0.5)^2\}.$$

The four effect functions are *constant* $\theta(z) = 1$, *linear* $-0.5 + 3z$, *threshold* $-0.5 + 2.5/(1 + e^{-12(z-0.5)})$ and *hump* $2 - 8(z - 0.5)^2$. The baseline sets $\rho = 1$; for the threshold design we vary the confounding strength ($\rho \in \{0.25, 0.5, 2\}$) and the number of countries ($G \in \{50, 200\}$). This design favours flexible estimators by construction, since the confounding is strongly nonlinear and the residual variation in adoption is larger than in the application; the plasmode design below addresses this.

We compare twelve estimators: (1) TWFE with a constant effect; (2) TWFE with $D \times Z$ and $D \times Z^2$; (3) TWFE-spline, with additive spline expansions of every control and a spline in Z for the effect; (4) FE-DML with a constant effect (Clarke and Polselli, 2026); (5) a boosted R-learner (Nie and Wager, 2021) fitted to the FE-DML residuals, with its hyperparameters chosen by country-level cross-validation; (6) a within-group DML sieve, which learns the nuisance functions on two-way demeaned data, in the spirit of the within-group estimator of Clarke and Polselli (2026); (7) a pooled DML sieve without Mundlak features or

within step; (8) an ablation with Mundlak features but no within step; (9) an ablation with the within step but no Mundlak features; (10) Panel-DOSE without penalty; (11) Panel-DOSE; and (12) a latent-information oracle that replaces the learned nuisance functions by the conditional means of D and Y given the controls *and the true country and year effects*. The oracle uses information that no feasible estimator has; it is not the population version of the nuisance problem that Panel-DOSE solves, whose features contain Mundlak means rather than α_i , and we use it only as a benchmark. All machine-learning estimators use LightGBM (Ke et al., 2017) with 600 trees, five country-level folds and two repetitions of the split with a pooled penalty; the sieve estimators use four interior knots. We replicate each design 100 times (50 in the sensitivity analysis) and report the integrated root-mean-squared error (IRMSE) over 17 grid points in $[0.1, 0.9]$, the integrated bias, the RMSE of the high-minus-low tercile contrast, and the coverage of nominal 95% intervals with Monte Carlo standard errors.

4.2. Results of the stylised design

Table 1 (and Figure C.2 in the supplementary material) reports accuracy, and Table C.7 in the supplementary material reports coverage. Linear fixed-effects models are heavily biased because the control function is nonlinear and correlated with adoption. The quadratic interaction does not help: its errors are identical in the constant, linear and hump designs because these effect functions lie in the span of its regressors, so the problem is the control function, not the flexibility of the effect. TWFE with spline controls removes most of the bias but keeps an integrated bias of about 0.23.

Among the feasible flexible estimators, Panel-DOSE has the lowest IRMSE in the three heterogeneous designs (0.15, 0.23 and 0.22 for the linear, threshold and hump designs), 20–47% below TWFE-spline and 15–45% below the tuned R-learner. The constant-effect FE-DML estimator is best when the effect is in fact constant. The ablations locate the gain in these designs. Dropping the Mundlak features multiplies the error by 2.1 to 2.9 (0.45–0.52), whereas dropping the within step raises it by 7–16% (0.17–0.25). Learning the nuisance on

demeaned data (the within-group sieve) performs poorly (0.70–0.92), because $g(X)$ minus its country mean is not a function of the demeaned controls when g is nonlinear. This is a structural limitation of that benchmark under a nonlinear control function, not evidence that its learner is weaker; similarly, the R-learner works with scalar transformed residuals rather than with demeaned interaction regressors. The comparison therefore shows the value of the Mundlak features for this estimand and these designs, not a general superiority over panel DML methods built for other estimands. The latent-information oracle has 16–22% lower IRMSE than Panel-DOSE, which indicates how much could be gained from better nuisance estimates in this design.

Coverage is below nominal in some cells. For Panel-DOSE, pointwise coverage is 0.81–0.91 and simultaneous coverage 0.55–0.85. It is worst in the threshold design, where the penalty smooths the steep part of the curve, but it is also below nominal when θ is constant and there is no smoothing bias (0.84 simultaneous), so the data-dependent penalty and the finite number of clusters contribute as well. The unpenalised sieve has pointwise coverage of 0.93–0.94 and simultaneous coverage of 0.83–0.88. Coverage of the smooth functionals is closer to nominal: 0.88–0.92 for the average effect, 0.90–0.97 for the tercile contrast and 0.88–0.94 for the BLP slope. With the oracle nuisance, average-effect coverage is 0.95–0.97. The sup- t constancy test rejects in 0.09 of replications when the effect is constant (0.07 without penalty), somewhat above the nominal 5%, and the linearity test in 0.06 when it is linear (0.05 without penalty); against the threshold and hump designs the constancy test rejects in 1.00 and 0.95. Intervals from the TWFE-interaction model cover the true tercile contrast in only 0.03–0.17 of replications, so its apparent precision is not informative when the control function is misspecified.

The sensitivity analysis (Table C.8 in the supplementary material) qualifies the ranking. With strong confounding ($\rho = 2$) or with 50 or 200 countries, Panel-DOSE remains the most accurate estimator. With mild confounding, TWFE-spline is as accurate or more accurate (IRMSE 0.17 against 0.22 at $\rho = 0.25$ and 0.20 against 0.23 at $\rho = 0.5$), so the gain from

Table 1: Stylised Monte Carlo design: accuracy ($G = 100$, $T = 24$, $\rho = 1$).

Estimator	IRMSE				Integrated bias				RMSE, high-low contrast			
	Const.	Lin.	Thr.	Hump	Const.	Lin.	Thr.	Hump	Const.	Lin.	Thr.	Hump
TWFE	0.63	0.94	1.15	1.05	0.63	0.57	0.60	0.96	0.00	1.41	2.07	0.17
TWFE-interaction	0.83	0.83	0.87	0.83	0.77	0.77	0.66	0.77	0.52	0.52	0.73	0.52
TWFE-spline	0.29	0.29	0.29	0.29	0.23	0.23	0.23	0.23	0.16	0.16	0.16	0.16
FE-DML (constant)	0.07	0.74	0.99	0.49	0.06	0.06	0.07	0.24	0.00	1.41	2.07	0.17
FE-DML + R-learner (tuned)	0.24	0.28	0.28	0.30	0.06	0.06	0.06	0.10	0.16	0.31	0.37	0.19
WG-DML sieve	0.61	0.74	0.70	0.92	0.53	0.45	0.48	0.81	0.40	0.78	0.68	0.31
Pooled DML sieve	0.63	0.64	0.68	0.71	0.60	0.61	0.63	0.66	0.27	0.29	0.30	0.28
Ablation: Mundlak only	0.18	0.17	0.25	0.25	0.10	0.07	0.07	0.14	0.21	0.23	0.25	0.21
Ablation: within only	0.44	0.45	0.49	0.52	0.41	0.41	0.42	0.46	0.24	0.27	0.28	0.25
Panel-DOSE (unpenalised)	0.26	0.26	0.27	0.26	0.05	0.06	0.07	0.05	0.20	0.22	0.22	0.20
Panel-DOSE	0.15	0.15	0.23	0.22	0.05	0.06	0.06	0.08	0.20	0.20	0.22	0.20
Latent-information oracle	0.12	0.12	0.20	0.18	0.00	0.00	-0.00	0.03	0.18	0.18	0.19	0.18

Notes: averages over 100 replications. The lowest IRMSE among feasible estimators in each design is in bold. The latent-information oracle uses the true country and year effects and is infeasible. IRMSE and integrated bias are computed over 17 grid points in $[0.1, 0.9]$.

machine learning is specific to strongly nonlinear confounding. Simultaneous coverage of the penalised bands falls to 0.40 at $\rho = 2$. The cross-validated penalty selects $\lambda = 0$ in 0% and the upper end of the grid (an effectively linear fit) in 33% of replications.

4.3. Plasmode design

The stylised design has less noise and more residual variation in adoption than the application. The plasmode design (Franklin et al., 2014) keeps the actual controls, treatment, moderators and panel structure of the baseline sample of Section 5 (4,075 observations, 149 countries) and regenerates only the outcome, as $Y_{it}^* = \theta(Z_{it})D_{it} + \hat{g}(F_{it}) + w_i\hat{e}_{it}$. Here $\hat{g}$ is a boosted-tree fit of observed productivity growth on the nuisance features, $\hat{e}_{it}$ are the corresponding residuals, and w_i are Rademacher weights drawn per country. The errors thus keep their size, heteroskedasticity and within-country dependence, while any association with D is randomised away. The design inherits the limitations of a single pilot fit: $\hat{g}$ is smoother than the true control function, the design holds the panel fixed, and the country-level sign flips remove any correlation of the errors across countries within a year beyond the year effects. For each of the three moderators we consider no effect (the size of the constancy test), a linear gradient (the size of the linearity test and the power of the constancy test) and a smooth threshold (curvature). The linear and threshold functions are scaled so that their

tercile contrast equals the minimum detectable contrast of the application, 2.8 times the baseline standard error (10.07, 9.16 and 31.27). The estimator is run as in the application’s robustness checks, with ten repetitions of the split and one cross-validated penalty pooled over the splits, and the pilot fit $\hat{g}$ is itself cross-fitted by country. We use 200 replications under no effect, so that the Monte Carlo standard error of a rejection rate of 5% is 1.5 percentage points, and 100 under each alternative.

Table 2 reports the results. Under no effect, the penalised constancy test rejects in 0.07, 0.09 and 0.06 of the replications for schooling, initial productivity and years since take-off, and the unpenalised test in 0.04, 0.07 and 0.03; with a Monte Carlo standard error of about 1.5 percentage points, the penalised test is slightly over-sized for initial productivity and close to nominal otherwise. The FE-DML test of a zero average association rejects in 0.03–0.06 of the replications, so the headline interval has close to nominal coverage in this design. Under the linear gradient, the penalised linearity test rejects in 0.04, 0.05 and 0.03 of the replications, and the constancy test detects a gradient of the size of the minimum detectable contrast in 0.93, 0.81 and 1.00; without the penalty the power is much lower along initial productivity (0.34). The contrast intervals cover the true contrast in 0.95, 0.96 and 0.94 of the replications under no effect, and the contrast estimates are biased by 0.88, 0.46 and 0.82, about a quarter of their standard errors; this second-order bias of the nuisance step is of the same order as the contrasts estimated in the application, which is another reason not to over-interpret small contrasts. The linearity test has little power against the threshold function (0.17, 0.04 and 0.38), and the simultaneous bands of the penalised fit cover it in 0.80, 0.87 and 0.79 of the replications, which reflects the smoothing bias described in Remark 1. The linear TWFE benchmarks are much less reliable along initial productivity (Table C.9 in the supplementary material): with no effect at all, the linear-interaction and spline-control contrasts are biased by 0.59 and 1.05 and reject in 0.11 and 0.33 of the replications.

Table 2: Plasmode design built on the empirical panel: Panel-DOSE.

True θ	High-low tercile contrast				Test rejection		Band coverage	
	Bias	RMSE	Coverage	Rejection	Constancy	Linearity	Pointwise	Simultaneous
<i>Moderator: Schooling</i>								
no effect	0.88	3.67	0.95 (0.02)	0.05 (0.02)	0.07 (0.02)	0.05 (0.02)	0.93 (0.01)	0.93 (0.02)
no effect, unpenalised	0.59	3.39	0.97 (0.01)	0.03 (0.01)	0.04 (0.01)	0.03 (0.01)	0.95 (0.01)	0.96 (0.01)
linear gradient	2.99	4.46	0.94 (0.02)	0.92 (0.03)	0.93 (0.03)	0.04 (0.02)	0.92 (0.01)	0.87 (0.03)
linear gradient, unpenalised	2.45	3.96	0.97 (0.02)	0.92 (0.03)	0.76 (0.04)	0.03 (0.02)	0.93 (0.01)	0.92 (0.03)
threshold	3.45	4.68	0.92 (0.03)	0.95 (0.02)	0.96 (0.02)	0.17 (0.04)	0.90 (0.01)	0.80 (0.04)
threshold, unpenalised	2.11	3.77	0.96 (0.02)	0.89 (0.03)	0.83 (0.04)	0.07 (0.03)	0.93 (0.01)	0.92 (0.03)
<i>Moderator: Initial productivity</i>								
no effect	0.46	3.15	0.96 (0.01)	0.04 (0.01)	0.09 (0.02)	0.06 (0.02)	0.94 (0.01)	0.94 (0.02)
no effect, unpenalised	0.68	4.49	0.92 (0.02)	0.09 (0.02)	0.07 (0.02)	0.07 (0.02)	0.95 (0.01)	0.94 (0.02)
linear gradient	0.90	3.35	0.98 (0.01)	0.85 (0.04)	0.81 (0.04)	0.05 (0.02)	0.93 (0.01)	0.96 (0.02)
linear gradient, unpenalised	1.30	4.65	0.93 (0.03)	0.60 (0.05)	0.34 (0.05)	0.05 (0.02)	0.95 (0.01)	0.96 (0.02)
threshold	0.38	3.16	0.95 (0.02)	0.81 (0.04)	0.75 (0.04)	0.04 (0.02)	0.92 (0.01)	0.87 (0.03)
threshold, unpenalised	1.38	4.74	0.92 (0.03)	0.61 (0.05)	0.42 (0.05)	0.06 (0.02)	0.96 (0.01)	0.96 (0.02)
<i>Moderator: Years since take-off</i>								
no effect	0.82	7.19	0.94 (0.02)	0.06 (0.02)	0.06 (0.02)	0.04 (0.01)	0.96 (0.01)	0.94 (0.02)
no effect, unpenalised	3.65	13.75	0.98 (0.01)	0.02 (0.01)	0.03 (0.01)	0.04 (0.01)	0.96 (0.01)	0.96 (0.01)
linear gradient	-1.60	6.91	0.98 (0.01)	0.97 (0.02)	1.00 (0.00)	0.03 (0.02)	0.96 (0.01)	0.97 (0.02)
linear gradient, unpenalised	-0.37	12.73	0.98 (0.01)	0.41 (0.05)	0.95 (0.02)	0.03 (0.02)	0.95 (0.01)	0.96 (0.02)
threshold	0.17	7.26	0.96 (0.02)	0.91 (0.03)	1.00 (0.00)	0.38 (0.05)	0.88 (0.02)	0.79 (0.04)
threshold, unpenalised	-0.17	12.69	0.96 (0.02)	0.44 (0.05)	0.97 (0.02)	0.18 (0.04)	0.96 (0.01)	0.95 (0.02)

Notes: Monte Carlo standard errors in parentheses. Under no effect, “Rejection” is the size of the 5% test that the contrast is zero and “Constancy” and “Linearity” are the sizes of the sup- t tests; under a linear gradient, “Linearity” is a size and the other rejection rates are powers. Band coverage refers to the true θ on a 17-point grid between the 5th and 95th percentiles of the moderator.

5. Empirical application: internet diffusion and productivity growth, 1996–2025

5.1. Data and specification

The main data set combines the World Bank’s World Development Indicators (WDI, July 2026 release, retrieved in September 2026) with the UNDP Human Development Report 2025. The outcome is annual growth of GDP per person employed at constant 2021 PPP prices, $Y_{it} = 100 \Delta \ln(\text{GDP}_{it}/\text{EMP}_{it})$, which WDI compiles from national accounts and the modelled employment estimates of the International Labour Organization (ILO) and which is available up to 2025. The treatment is the share of the population using the internet (ITU) in year $t - 1$, measured on $[0, 1]$. It is a lagged *level*, not a change: θ compares growth in years in which a country’s internet share was higher with growth in years in which it was lower, conditional on the controls and on country and year effects, and one tenth of θ is the difference in growth (in pp) associated with a ten-percentage-point higher lagged share. In the baseline, the controls are lagged mean years of schooling, the lagged government

consumption share, trade openness, population growth, the age-dependency ratio and the urban population share, and initial log GDP per worker, measured as the average over 1991–1995.

The baseline deliberately excludes three groups of variables used in the earlier literature. Lagged productivity levels and growth are excluded because they violate strict exogeneity mechanically (Section 3). Mobile and fixed-broadband subscriptions are excluded because they measure the same diffusion process, so that controlling for them turns the estimand into the association of internet use net of co-diffusing technologies. The investment share is excluded because it is a likely channel. We report specifications that add each group back. Variables that enter only such robustness checks do not restrict the baseline sample: an observation is used if the outcome, the treatment and the baseline controls are observed. The sample covers 1996–2025, drops growth observations below the 1st and above the 99th percentile, and keeps countries with at least 15 usable years. Table B.1 in Appendix B shows the observations lost at each step. We screen the adoption series for implausible values and jumps; the rules flag 39 values, for example a rise in Fiji’s mobile subscriptions from 361 to 574 per 100 people in one year (Appendix B). Flagged values are set to missing, so flags in the mobile and broadband series affect only the specifications that use these series. The resulting panel contains 4,075 observations on 149 countries. It includes 53 of 86 high-income, 45 of 59 upper-middle-income, 37 of 47 lower-middle-income and 14 of 25 low-income economies. Economies are excluded mainly because productivity, schooling or government-consumption data are missing; many of the excluded low-income economies are fragile or conflict-affected states (e.g. Afghanistan, Somalia, South Sudan, Yemen). A stricter sample that also requires the other ICT series, the investment share, lagged productivity and a second lag of the treatment to be observed—the complete-ICT sample (3,787 observations, 141 countries)—is used as a robustness check. Table C.3 in the supplementary material reports descriptive statistics.

We also use two releases of the Penn World Table (Feenstra et al., 2015): version 11.0

(Groningen Growth and Development Centre, 2025), which covers 118 countries over 1996–2023 in our sample (3,178 observations), and version 10.0, which ends in 2019 (115 countries, 2,658 observations). The outcome is growth of real GDP at constant national prices per person engaged (rgdpna/emp), the variable PWT recommends for growth comparisons. The initial-productivity moderator is instead output-side real GDP at current PPPs per person engaged (cgdp/emp), averaged over 1991–1995, which PWT recommends for comparing productive capacity across countries at a point in time; we report the national-prices level as a sensitivity check. PWT 10.0 expresses levels in 2017 and PWT 11.0 in 2021 US dollars. Because the knots and terciles are placed at quantiles and the B-spline basis is invariant to affine transformations of the moderator, a change of base year does not by itself affect the estimates; what matters is how the releases rank countries. The PWT panels use the PWT human-capital index and the same controls as the WDI baseline plus the labour share; PWT also provides total factor productivity (TFP).

We estimate model (1) with three moderators corresponding to the predictions of Section 2.5: mean years of schooling (absorptive capacity), initial log GDP per worker (distance to the frontier and leapfrogging) and years since the internet-user share first reached 10% (the J-curve). Using initial rather than lagged productivity as the development moderator avoids the mechanical link with the outcome and sorts countries rather than country-years into terciles. Years since take-off is computed from the whole adoption path, so before take-off its (negative) value uses the future year in which the share crosses 10%; it describes the stage of the diffusion life cycle but is not predetermined. We therefore also use a predetermined version that is zero until take-off and counts the years since take-off afterwards. Panel-DOSE uses LightGBM with 400 trees, five country-level folds, 20 repetitions of the split with one cross-validated penalty pooled over the splits, and cubic B-splines with four interior knots; standard errors are clustered by country. Robustness checks use ten repetitions of the split to save computation, so their FE-DML estimates can differ slightly from the baseline even when the specification is otherwise the same.

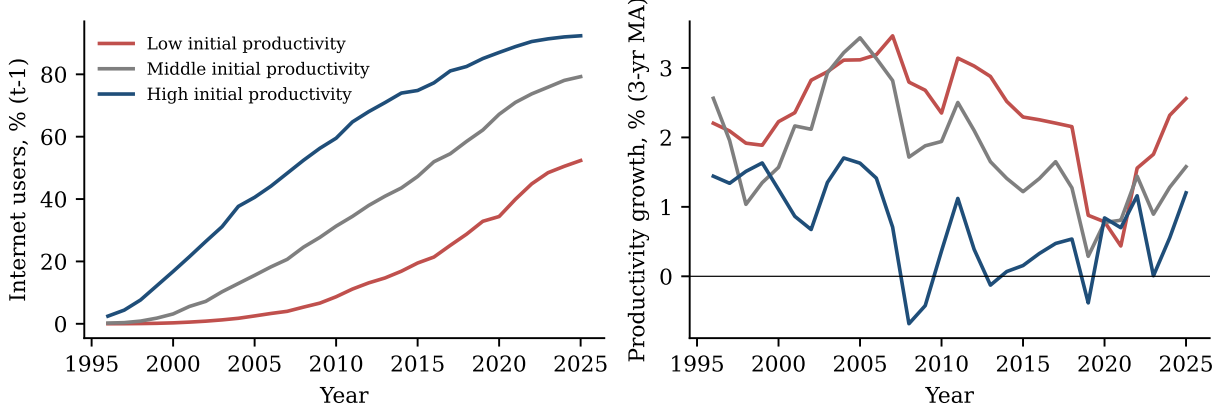


Figure 1: Internet diffusion and labour-productivity growth by tertile of initial GDP per worker, 1996–2025: (left) sample average of the lagged internet-user share by year; (right) centred three-year moving average of the yearly sample averages of productivity growth.

Figure 1 shows the variation that identifies the estimates. Internet use diffused everywhere, but at different times. In the countries with the highest initial productivity, about half of the population was online by the mid-2000s, whereas in the least productive third the share passed one fifth only in the mid-2010s and one half at the end of the sample. Smoothed productivity growth was highest in the low and middle groups during the 2000s and fell around 2009 and 2020. Year effects absorb these common shocks, so the estimates rely on differences across countries in the timing of adoption relative to their own productivity paths.

5.2. The average association

In the baseline, the headline FE-DML estimate is -2.64 (s.e. 1.33): a ten-percentage-point higher lagged internet-user share is associated with -0.264 pp lower productivity growth, with a country-clustered 95% interval of $[-0.524, -0.004]$ pp. This interval only just excludes zero, and the conclusion is not robust to reasonable choices. Two-way clustering by country and year gives a standard error of 1.40 and an interval of $[-0.538, 0.009]$ pp. Across 20 random seeds for the sample splits, the estimate ranges from -2.81 to -2.32 (mean -2.61), and the country-clustered interval excludes zero for 65% of the seeds. We therefore read the flexible estimate as negative but not clearly different from zero.

The linear TWFE estimate with the same controls is -2.84 (0.79), and it is stable across checks for trending and convergence-type confounding. It is -3.71 (0.65) with country-specific linear trends, -4.43 (1.24) with quadratic trends, -2.73 (0.74) when lagged log productivity is added, -2.74 (1.03) with year effects specific to terciles of initial productivity, and -2.26 (0.88) with region-by-year effects (Table C.4 in the supplementary material). Mean reversion along a convergence path therefore does not explain the negative sign. The FE-DML estimate is less precise than TWFE because the nuisance step removes most of the variation in adoption. Adoption is highly predictable from the controls, their country means and the year: the out-of-fold R^2 of the treatment nuisance is 0.96 (0.77 within countries), against 0.20 (0.03) for the outcome (Table C.5 in the supplementary material). Results are similar across learning rates, tree sizes and numbers of trees, with FE-DML estimates between -2.94 and -2.23 . After the within transformation, only 0.22 of the within-country variance of adoption remains (2.5% of its total variance), and only 0.11–0.14 in the lowest tercile of schooling or initial productivity, where the median adoption share is 0.04–0.07.

The sign of the FE-DML estimate is negative in 32 of the 32 specifications with the baseline outcome and treatment, but only 17 of their country-clustered intervals exclude zero (Figure C.1 in the supplementary material). Controlling for mobile and broadband subscriptions reduces the estimate to -0.50 (1.24). These specifications estimate a different quantity—the association of internet use net of co-diffusing technologies—so the fall shows how much of the baseline association the three adoption series share. The estimate is also smaller without the transition economies of Eastern Europe and Central Asia (-1.56 , s.e. 1.29), whose 1991–1995 productivity reflects the post-socialist collapse and whose recovery growth slowed while internet use rose, and without the 8 mainly conflict-affected or fuel-exporting economies that are outside the complete-ICT sample (-2.41 , 1.25). The association is concentrated in the first half of the period. It is -4.91 (2.18) over 1996–2011 and -0.70 (2.15) over 2012–2025; with TWFE it is -4.35 (1.01) and 0.70 (1.46). The 1996–2011 period combines the early diffusion of the internet with the catch-up and commodity boom

of the 2000s, whose effects may differ by level of development in ways that the controls do not capture.

The negative association is in output rather than in employment. With TWFE, the lagged internet share is associated with -3.83 (0.78) lower GDP growth and -0.99 (0.33) lower employment growth, so a larger employment denominator does not explain the lower growth of output per worker.

Exogeneity and dynamics. Strict exogeneity implies that future adoption should not predict current growth once current adoption is controlled for (Wooldridge, 2010, Section 10.7). Adding the lead $D_{i,t+1}$ to the baseline gives a lead coefficient of -0.87 (1.85) with TWFE and -1.14 (2.06) with FE-DML (Table 3). Neither is significant, but the test has little power because adoption is highly persistent. Used alone, the lead is associated with growth as strongly as the lag (FE-DML -3.63 , s.e. 1.29). In a TWFE profile that dates the share from $t - 6$ to $t + 4$, the coefficients peak around $t - 1$ and t (-3.24 and -2.92) and decline in both directions (-1.81 at $t - 6$, -1.18 at $t + 4$). The association is therefore a low-frequency co-movement between the level of adoption and growth, not a response to changes in adoption: the lagged change $D_t - D_{t-1}$ is associated with growth by -0.37 (1.84) with TWFE and -2.30 (1.76) with FE-DML.

Table 3 and Figure 3(b) report local projections (Jordà, 2005) of the cumulative change in log productivity from $t - 1$ to $t + h$ on the internet share in $t - 1$, with the baseline controls; horizon h covers $h + 1$ years of growth. With a persistent regressor, these are close to the cumulated annual association and do not add independent evidence. They show -0.58 pp per ten points over two years ($h = 1$), -1.16 pp over four years and -1.79 pp over nine years, with intervals that include zero from $h = 4$ onwards; on the 2,941 observations for which the nine-year change is observed the pattern is the same. These regressions describe how the conditional association evolves; they do not identify an impulse response to an adoption shock. Within the horizons we observe, they show no delayed positive association. Non-overlapping five-year averages (Islam, 1995) give -2.43 (0.84) with TWFE and -2.59

(1.33) with FE-DML.

The picture is different across countries. We regress average annual productivity growth over 2000–2024, i.e. $100(\ln LP_{2024} - \ln LP_{1999})/25$, which is the mean of the annual growth rates used in the panel, on the change in the panel treatment over the same years, $D_{2024} - D_{2000}$ (the internet share of 2023 minus that of 1999), with the controls of 2000 (mostly 1999 values). Economies whose internet share rose more grew faster, by about 0.22 pp per year per ten points (OLS 2.21, s.e. 0.62; DML 2.17, s.e. 1.14; 117 countries). Built from productivity levels without the panel’s trimming and minimum-length rules, the long difference covers 122 countries and gives 2.58 (0.62). Because this comparison does not remove country effects, it is consistent with faster-growing economies adopting faster rather than with adoption raising growth. An instrumental-variable check that instruments adoption with the 1990 density of fixed telephone lines interacted with the global diffusion curve has a weak first stage ($F = 5.6$) and gives an imprecise two-stage least squares (2SLS) estimate of -3.78 (4.00); we report it for completeness only.

Comparison with broadband studies. Our main estimate concerns internet users, per-worker growth and a global panel, whereas Czernich et al. (2011) study broadband, per-capita growth and OECD countries over 1996–2007 with an instrumental-variable design. As a descriptive benchmark closer to their setting, we use fixed broadband subscriptions as the treatment and GDP-per-capita growth as the outcome in high-income economies. The FE-DML estimate is -2.39 (2.55), i.e. $[-0.74, 0.26]$ pp per ten points. Because the populations, periods, treatments and estimands differ, and ours rests on selection on observables, this comparison cannot establish whether the two sets of results are compatible; it shows only that a conditional association of the size they report is not visible in these data. In all countries, the same specification gives -6.35 (2.52), a negative association comparable to that of internet use.

Table 3: Dynamic specifications and alternative designs.

<i>Panel A. Local projections: change in log productivity ($\times 100$) from $t - 1$ to $t + h$</i>							
Horizon (years)	All available observations			Common sample		High – low	
	Average	(s.e.)	Obs.	Average	(s.e.)	Contrast	(s.e.)
$h = 0$ (1)	−2.71**	(1.29)	4,075	−4.31***	(1.59)	−0.67	(3.23)
$h = 1$ (2)	−5.79**	(2.34)	3,933	−8.85***	(3.04)	0.14	(5.68)
$h = 2$ (3)	−7.84**	(3.70)	3,792	−10.63**	(4.63)	4.01	(9.01)
$h = 3$ (4)	−11.58**	(5.43)	3,649	−12.38**	(6.04)	7.04	(13.88)
$h = 4$ (5)	−11.82*	(6.93)	3,508	−12.33*	(7.49)	13.98	(18.46)
$h = 5$ (6)	−12.66	(8.25)	3,370	−12.86	(8.75)	24.66	(21.98)
$h = 6$ (7)	−11.06	(9.13)	3,230	−14.07	(9.44)	35.31	(28.66)
$h = 7$ (8)	−14.58	(10.43)	3,087	−15.03	(10.26)	42.04	(33.52)
$h = 8$ (9)	−17.87*	(10.71)	2,941	−17.87*	(10.71)	47.33	(37.57)
<i>Panel B. Alternative designs</i>							
Specification	Estimate	(s.e.)	Obs.				
Five-year averages, TWFE	−2.43***	(0.84)	834				
Five-year averages, FE-DML	−2.59*	(1.33)	834				
Long difference, OLS	2.21***	(0.62)	117				
Long difference, DML	2.17*	(1.14)	117				
Lead D_{t+1} , TWFE	−0.87	(1.85)	3,923				
Lead D_{t+1} , FE-DML	−1.14	(2.06)	3,923				
Fixed-line IV (2SLS), first-stage $F = 5.6$	−3.78	(4.00)	3,977				
Broadband, GDP p.c. growth, high income	−2.39	(2.55)	1,545				
Broadband, GDP p.c. growth, all countries	−6.35**	(2.52)	3,995				

Notes: coefficients per unit of adoption (0–1); divide by 10 for a ten-point difference. Panel A: FE-DML average and Panel-DOSE tercile contrast by initial log GDP per worker, baseline controls, ten repetitions of the split (so the $h = 0$ average differs slightly from the 20-split baseline); the horizon in parentheses is the number of annual growth rates covered, $h + 1$. “Common sample”: observations for which the nine-year change is observed. Five-year averages: non-overlapping periods 1996–2000 to 2021–2025, treatment and controls at the start of the period. Long difference: average annual growth of GDP per worker over 2000–2024 on $D_{2024} - D_{2000}$ (internet share 2023 minus 1999), with 2000 values of the controls; OLS with heteroskedasticity-robust (HC1) standard errors and cross-fitted partially linear DML. Lead: coefficient on D_{t+1} when it is added to the baseline (strict-exogeneity check). IV: 2SLS with country and year effects; instrument = 1990 fixed-line density $\times$ leave-one-out global internet share. Broadband rows: FE-DML with fixed broadband subscriptions as treatment and GDP-per-capita growth as outcome. *, **, ***: 10%, 5%, 1%.

5.3. Heterogeneity

Table 4 and Figure 2 report the heterogeneity estimates. Table C.10 in Appendix C reports the selected penalties, effective degrees of freedom, identifying variation and treatment support by tercile. The bands in Figure 2 describe the penalised fit (Remark 1); the tests of heterogeneity are the constancy tests.

Schooling. The association does not vary with schooling. The estimated function is nearly flat (Figure 2a), the high-minus-low tercile contrast is -0.70 (s.e. 3.60), the BLP slope is -0.10 (0.49) per year of schooling, and the constancy test does not reject ($p = 0.96$). The

absorptive-capacity prediction of a positive gradient is therefore not supported, but the data cannot rule out moderate gradients either: the 95% interval for the contrast is $[-0.78, 0.64]$ pp per ten points, and the minimum detectable contrast (80% power at 5%) is about 1.01 pp per ten points. A null result for heterogeneity is not evidence against complementarities of this size. The sign of the contrast varies across robustness checks; in the complete-ICT sample it is -3.55 (3.36), and over 1996–2011 it is -12.61 (5.75, constancy $p = 0.06$), against 9.29 (5.08) over 2012–2025. Learning-adjusted years of schooling, which account for the quality of schooling (Angrist et al., 2021), give a similar picture (contrast 0.44, s.e. 3.46).

Initial productivity. Along initial productivity, the WDI estimates show no monotone gradient. The contrast is -0.76 (3.27) and the constancy test gives $p = 0.16$. The estimated function dips in the middle of the distribution (Figure 2b), and in some checks the constancy test rejects at 5%: without the penalty ($p = 0.04$), in the period 1996–2011 ($p = 0.02$, contrast 13.10, s.e. 8.05), without Mundlak features and in a few others in Appendix C. These fits select a small penalty and use four to eight effective degrees of freedom; the penalised test is slightly over-sized for this moderator in the plasmode (0.09 at the 5% level, Section 4.3), and the tercile contrasts of these checks have different signs. We read them as weak evidence of non-monotone variation, not as support for the monotone gradient that the leapfrogging or frontier arguments predict. In the local projections the contrast grows with the horizon, to 47.33 (37.57) at $h = 8$, but it is imprecise at every horizon (Table 3). Excluding the bottom 5% of the moderator gives a contrast of 0.40 (3.15), and using only countries with pre-sample productivity data gives 0.00 (3.20). Distance to the US frontier as the moderator, with terciles formed within year, gives -5.16 (2.88); because this moderator is built from lagged productivity, it is not predetermined, and we report it as descriptive only. The linear benchmarks find a positive gradient: 1.77 (0.85) with a linear interaction and 3.90 (1.14) with spline controls. These values are close to the biases of the same estimators in the plasmode design without any effect (0.59 and 1.05), so they are what restrictive control functions produce in these data rather than evidence of a gradient.

Years since take-off. The J-curve prediction receives no statistical support. In the baseline, the contrast between the last and the first tercile of time since take-off is -0.11 (11.17) and the constancy test does not reject ($p = 0.09$); the first tercile contains the years long before take-off in countries that adopted late, where the estimate is very imprecise. With the predetermined moderator, which is zero before take-off, the contrast is 4.95 (3.02, $p = 0.19$), and in the complete-ICT sample it is 5.72 (4.88). The more negative association in the first years after take-off that appears in the complete-ICT sample is therefore not robust to the choice of sample.

Dose response. The marginal association $\hat{f}'(d)$ is negative over the range of adoption, most clearly at low adoption (Figure 3a), and linearity of f is not rejected ($p = 0.14$). We do not detect departures from a linear conditional dose response over the evaluated support, but the linearity test has little power against curvature of the size considered in the plasmode (Section 4.3), so the absence of visible network effects or saturation is weak evidence.

Multiple testing. The three primary constancy tests give $p = 0.96$, 0.16 and 0.09 , with Holm-adjusted p -values of 0.96 , 0.33 and 0.28 . Without the penalty, the tests give $p = 0.57$, 0.04 and 0.03 (Holm 0.57 , 0.10 , 0.10); the plasmode shows the unpenalised tests to have close to nominal size but low power.

5.4. Data sources and vintages

The PWT panels tell a different story about heterogeneity (Panel B of Table 4 and Figure 4b). With the same estimator, the contrast along initial productivity is 14.42 (4.92) in PWT 10.0 (1996–2019) and 14.68 (3.88) in PWT 11.0 (1996–2023), and the constancy test rejects in both ($p = 0.008$ and $p < 0.001$), against 2.01 (3.55, $p = 0.61$) in the WDI data for the countries and years of the PWT 11.0 panel. The PWT gradient comes from a strongly negative association in the least productive economies. It is smaller when initial productivity is measured at national prices rather than at current PPPs (12.21 in PWT

Table 4: Heterogeneous associations of internet diffusion with labour-productivity growth.

Specification	Average	Tercile of moderator			High −	BLP	Constancy
	(FE-DML)	Low	Middle	High	low	slope	
<i>Panel A. Moderator: mean years of schooling</i>							
Baseline	−2.64** (1.33)	−2.15 (3.00)	−2.63* (1.44)	−2.72* (1.65)	−0.70 (3.60)	−0.10 (0.49)	0.96
Complete-ICT sample	−2.44** (1.22)	0.21 (2.63)	−2.75** (1.28)	−3.25* (1.68)	−3.55 (3.36)	−0.51 (0.46)	0.40
Country-specific linear trends [†]	−3.00** (1.38)	−5.96** (3.03)	−1.62 (2.01)	−0.66 (1.67)	4.99 (3.50)	0.46 (0.47)	0.22
Excluding transition economies [†]	−1.56 (1.29)	−3.07 (2.86)	−1.83 (1.53)	−0.68 (1.34)	2.22 (3.32)	0.30 (0.45)	0.47
Period 1996–2011 [†]	−4.91** (2.18)	5.99 (4.87)	−2.42 (3.03)	−6.73*** (2.39)	−12.61** (5.75)	−1.76** (0.79)	0.06
Period 2012–2025 [†]	−0.70 (2.15)	−5.06 (3.70)	0.54 (1.88)	4.19 (2.68)	9.29* (5.08)	1.29* (0.70)	0.09
Adding mobile and broadband [†]	−0.50 (1.24)	−1.40 (2.41)	−0.60 (1.24)	0.10 (1.60)	1.62 (3.02)	0.22 (0.41)	0.54
PWT 11.0, 1996–2023	−2.60* (1.33)	−11.56* (6.07)	−1.58 (1.86)	−0.82 (1.24)	11.14* (6.19)	9.64* (5.02)	0.22
PWT 10.0, 1996–2019	−2.12 (1.54)	−10.32* (5.69)	−0.95 (2.13)	−2.12* (1.09)	8.00 (5.87)	9.10** (4.49)	0.04
<i>Panel B. Moderator: initial log GDP per worker</i>							
Baseline	−2.64** (1.33)	−0.59 (3.00)	−4.79*** (1.81)	−1.27 (1.43)	−0.76 (3.27)	−0.81 (1.40)	0.16
Complete-ICT sample	−2.44** (1.22)	−1.69 (2.71)	−2.35* (1.30)	−3.05** (1.34)	−1.10 (3.18)	−0.44 (1.27)	0.68
Country-specific linear trends [†]	−3.00** (1.38)	−4.79 (3.00)	−3.23** (1.48)	−1.73 (1.48)	2.85 (3.48)	1.14 (1.39)	0.42
Excluding transition economies [†]	−1.56 (1.29)	5.16* (2.96)	−4.88** (2.35)	0.31 (1.46)	−4.83 (3.18)	−3.31*** (1.27)	0.06
Period 1996–2011 [†]	−4.91** (2.18)	−12.98 (8.07)	−7.89** (3.66)	−1.19 (1.88)	13.10 (8.05)	3.51 (3.55)	0.02
Period 2012–2025 [†]	−0.70 (2.15)	−1.54 (2.97)	−0.54 (2.17)	0.27 (3.02)	1.99 (4.32)	0.79 (1.71)	0.54
WDI data, PWT 11.0 countries, 1996–2023	−3.80*** (1.30)	−5.17* (2.69)	−3.95*** (1.32)	−3.12* (1.79)	2.01 (3.55)	0.88 (1.56)	0.61
PWT 11.0, 1996–2023	−2.60* (1.33)	−12.09*** (3.52)	−3.66** (1.62)	2.46* (1.42)	14.68*** (3.88)	7.26*** (2.07)	<0.01
PWT 10.0, 1996–2019	−2.12 (1.54)	−12.19*** (3.83)	−3.28** (1.49)	2.14 (1.83)	14.42*** (4.92)	6.50*** (2.22)	0.01
<i>Panel C. Moderator: years since internet take-off (share ≥ 10%)</i>							
Baseline	−1.94 (1.38)	−0.26 (11.14)	−3.18* (1.67)	−0.74 (1.63)	−0.11 (11.17)	−0.28 (0.57)	0.09
Complete-ICT sample	−1.86 (1.31)	−6.41 (4.19)	−2.54 (1.59)	−0.73 (1.72)	5.72 (4.88)	0.24 (0.22)	0.49
Predetermined moderator (zero before take-off) [†]	−1.75 (1.38)	−4.62** (2.34)	−2.14 (1.54)	0.05 (1.76)	4.95 (3.02)	0.25 (0.18)	0.19

Notes: Panel-DOSE estimates; standard errors clustered by country in parentheses. Coefficients are per unit of adoption (0–1); divide by 10 for a ten-point difference. The dependent variable is annual labour-productivity growth (%) unless stated otherwise. Baseline: WDI panel (149 countries, 1996–2025) with the controls listed in Section 5. “Complete-ICT sample”: observations with every ICT series, the investment share, lagged productivity and the second lag of the treatment observed (141 countries). “Country-specific linear trends” removes a linear trend for each country in the within step. “Excluding transition economies” drops the post-socialist economies of Eastern Europe and Central Asia. PWT rows use the Penn World Table panels, with the PWT human-capital index (Panel A) and initial log PWT labour productivity at current PPPs (Panel B) as moderators. The BLP slope is per unit of the moderator (years of schooling, log points, years). “Constancy test p ”: sup- t test that θ is constant. [†]: ten repetitions of the sample split (20 otherwise), one penalty pooled over the splits. The high-minus-low contrast is aggregated across splits as a functional in its own right, so it can differ slightly from the difference of the two tercile columns. *, **, ***: 10%, 5%, 1%.

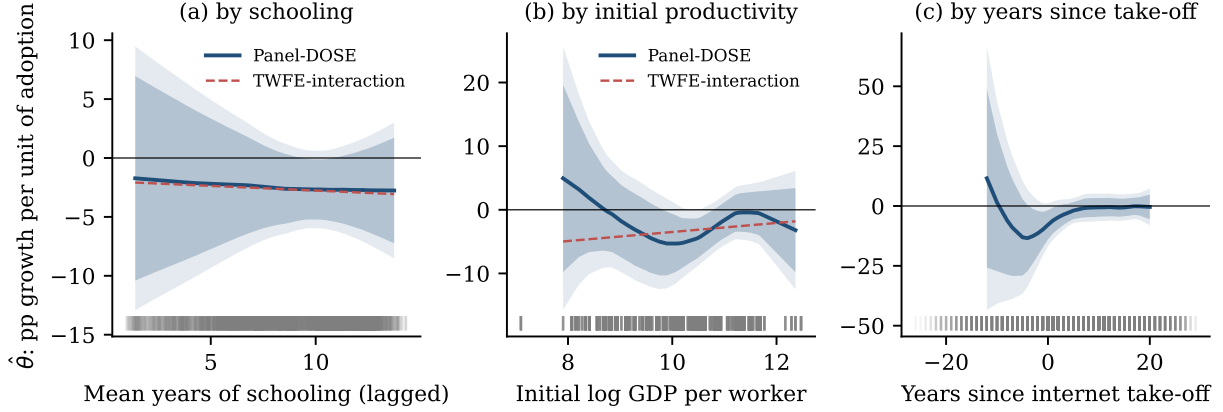


Figure 2: Estimated association of lagged internet diffusion with labour-productivity growth, WDI panel 1996–2025: (a) by mean years of schooling and (b) by initial log GDP per worker, each with the TWFE linear-interaction estimate (dashed); (c) by years since the internet-user share first reached 10%. Dark bands: 95% pointwise; light bands: 95% simultaneous, both for the penalised fit (Remark 1). Grids span the 1st to 99th percentiles and the vertical axes show the full bands; tick marks show the distribution of the moderator.

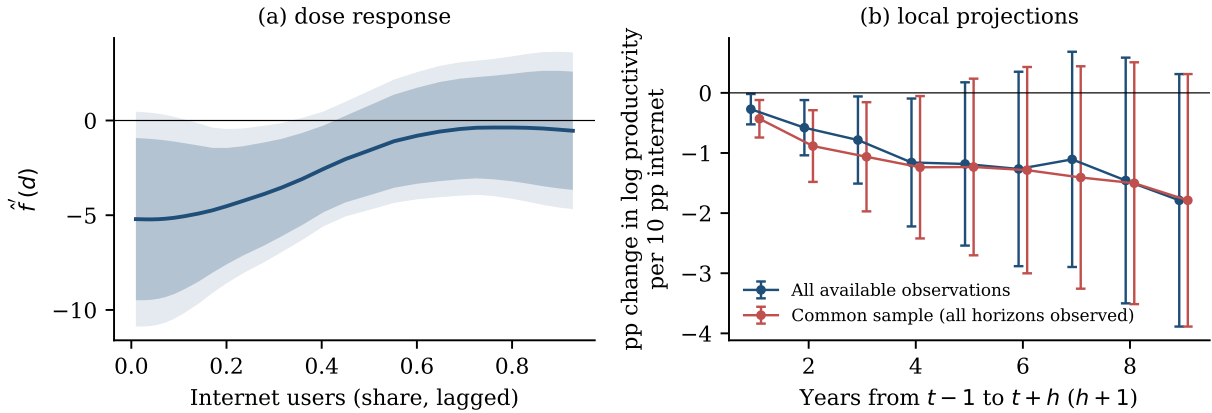


Figure 3: (a) Marginal association $\hat{f}'(d)$ in the dose-response model, from the 13th to the 95th percentile of adoption, with 95% pointwise and simultaneous bands. (b) Local projections: change in log labour productivity ($\times 100$) from $t - 1$ to $t + h$ associated with a ten-point higher internet share in $t - 1$, plotted against the number of years covered ($h + 1$), for all available observations and for the common sample in which the nine-year change is observed, with 95% intervals.

10.0 and 8.21 in PWT 11.0), and in PWT 11.0 restricted to 1996–2019 (7.73, s.e. 4.88). The linear benchmarks also find larger gradients in PWT than in WDI (TWFE with spline controls: 6.93 in PWT 11.0 against 3.90 in WDI), but the plasmode shows that they produce gradients of this size without any true effect.

These comparisons differ in more than the productivity series: the PWT panels include countries that are not in the WDI panel, and their controls include the PWT human-capital

index and the labour share instead of the dependency ratio and the urban share. Panel A of Table 5 therefore compares the two sources on the 2,839 country-years (108 countries) that both contain, replacing one component at a time and refitting each of the eight combinations. With every component from WDI the contrast is 1.89 (3.57); with every component from PWT 11.0 it is 9.29 (3.71). Because the components interact, we attribute the change with Shapley values, which average each component’s marginal contribution over all orders of replacement: the growth series accounts for 51% of the change, the control set for 38% and the moderator for 10%. The two sources agree on the ranking of countries (rank correlation of initial productivity 0.948), but their annual growth rates correlate at only 0.81, and at 0.75 after removing country and year effects—the variation the estimators use—with the weakest agreement in the least productive tercile (0.68 against 0.73 in the most productive; Table C.6 in the supplementary material). The difference between the sources is stable across split seeds (from -8.58 to -7.57) and to dropping any single country, and holding the smoother fixed gives similar values (1.82 and 9.98). It is, however, imprecise: a paired country bootstrap that refits both sources on the same resampled countries gives a standard error of 4.61 and a 95% interval of $[-15.44, 1.65]$. The data therefore suggest, but do not establish at conventional levels, that heterogeneity estimates differ between the sources, and about half of the difference comes from the growth series.

Panel B makes the same comparison for the two PWT releases on the 2,655 country-years (115 countries) they have in common. The gradient is large in both: 15.09 (4.91) with every component from PWT 10.0 and 12.90 (5.55) with every component from PWT 11.0. The difference of 2.19 has a bootstrap standard error of 4.94, varies between 1.28 and 6.69 across split seeds and between -0.29 and 5.55 when single countries are dropped, and its Shapley attribution changes sign between components. The two releases rank countries almost identically (rank correlation 0.996), and their growth rates correlate at 0.89. We therefore find no evidence that the revision from PWT 10.0 to PWT 11.0 changed the gradient; the revision matters much less for our question than the choice between PWT and

Table 5: Heterogeneity along initial productivity across data sources, on common country-years: component replacement.

Source of			Refitted			Fixed smoother	
Outcome	Moderator	Controls	High – low	(s.e.)	$\hat{\lambda}$	High – low	(s.e.)
<i>Panel A. WDI and PWT 11.0 (2,839 country-years, 108 countries, 1996–2023)</i>							
WDI	WDI	WDI	1.89	(3.57)	10^4	1.82	(3.26)
WDI	WDI	PWT 11.0	3.63	(3.52)	10^4	3.87	(3.25)
WDI	PWT 11.0	WDI	1.91	(3.42)	10^4	2.37	(3.42)
WDI	PWT 11.0	PWT 11.0	4.51	(3.36)	10^4	4.86	(3.54)
PWT 11.0	WDI	WDI	4.63	(3.02)	10^4	4.57	(3.10)
PWT 11.0	WDI	PWT 11.0	7.86**	(3.40)	17.8	8.18**	(3.25)
PWT 11.0	PWT 11.0	WDI	5.47	(3.47)	10^4	5.83*	(3.37)
PWT 11.0	PWT 11.0	PWT 11.0	9.29**	(3.71)	10^4	9.98***	(3.61)
<i>Shapley attribution of the change (PWT 11.0 – WDI)</i>							
Outcome			3.81			3.92	
Moderator			0.77			1.16	
Controls			2.82			3.08	
Total			7.40			8.16	
Difference WDI – PWT 11.0: –7.40; paired bootstrap s.e. 4.61, 95% interval [–15.44, 1.65]							
Across 10 split seeds: difference from –8.58 to –7.57; leave-one-country-out: –11.04 to –6.76							
<i>Panel B. PWT 10.0 and PWT 11.0 (2,655 country-years, 115 countries, 1996–2019)</i>							
PWT 10.0	PWT 10.0	PWT 10.0	15.09***	(4.91)	10^3	17.10***	(5.51)
PWT 10.0	PWT 10.0	PWT 11.0	12.78***	(4.95)	1,778	15.24***	(5.59)
PWT 10.0	PWT 11.0	PWT 10.0	15.74***	(5.47)	0.562	15.20***	(5.35)
PWT 10.0	PWT 11.0	PWT 11.0	14.94***	(5.43)	0.562	13.69**	(5.54)
PWT 11.0	PWT 10.0	PWT 10.0	12.70**	(5.35)	10^2	14.09***	(5.37)
PWT 11.0	PWT 10.0	PWT 11.0	8.91	(5.44)	10^4	11.14*	(5.93)
PWT 11.0	PWT 11.0	PWT 10.0	13.73***	(5.18)	1	12.82**	(5.45)
PWT 11.0	PWT 11.0	PWT 11.0	12.90**	(5.55)	0.562	11.48*	(5.97)
<i>Shapley attribution of the change (PWT 11.0 – PWT 10.0)</i>							
Outcome			–2.45			–2.82	
Moderator			2.08			–0.99	
Controls			–1.81			–1.81	
Total			–2.19			–5.61	
Difference PWT 10.0 – PWT 11.0: 2.19; paired bootstrap s.e. 4.94, 95% interval [–10.36, 10.57]							
Across 10 split seeds: difference from 1.28 to 6.69; leave-one-country-out: –0.29 to 5.55							

Notes: Panel-DOSE tercile contrast along initial log labour productivity on the country-years present in both sources; the treatment is the same ITU series in all rows. Each row takes the outcome, the moderator and the control set from the source indicated (WDI controls: schooling, government share, openness, population growth, dependency ratio, urban share; PWT controls: human-capital index, government share, openness, population growth, labour share). “Refitted”: nuisance functions, cross-validated penalty and tercile groups re-estimated for each configuration, 20 splits; standard errors conditional on the selected penalty. “Fixed smoother”: moderator transformed to within-source percentile ranks, tercile groups of the second source and one penalty for every row. Shapley values average the marginal contribution of each component over the six orders of replacement and sum to the total change. Paired country bootstrap: 199 draws of countries with replacement, both sources refitted on each draw (ten splits, copies of a country kept in the same fold); basic (bias-corrected) percentile interval. Split seeds: ten seeds of the two all-component fits; leave-one-country-out: both sources refitted without each country, with penalty, seed and tercile cutoffs fixed. *, **, ***: 10%, 5%, 1%.

World Bank data.

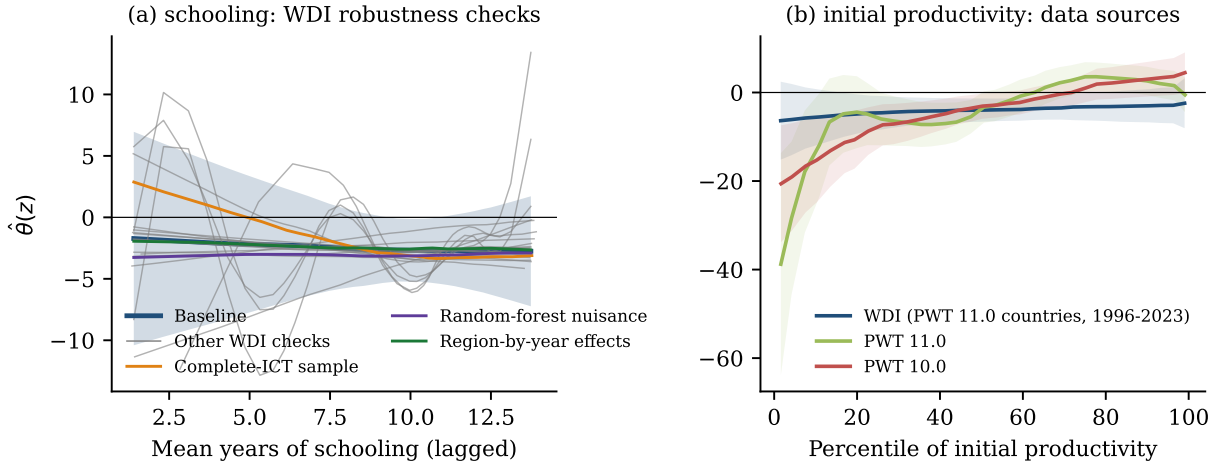


Figure 4: (a) $\hat{\theta}(z)$ by schooling across the WDI robustness checks that keep the baseline moderator, treatment and outcome (listed in Tables 4 and C.11); three checks are highlighted and the shaded band is the 95% pointwise band of the baseline. (b) $\hat{\theta}(z)$ by initial productivity in the WDI baseline, PWT 11.0 (1996–2023) and PWT 10.0 (1996–2019), with 95% pointwise bands; the horizontal axis is the percentile of the moderator within each sample, because the levels are measured in different units (constant 2021 PPP dollars in WDI, current PPPs in 2021 and 2017 US dollars in PWT 11.0 and 10.0).

6. Discussion

6.1. Interpretation

Three features of the results deserve emphasis. First, within countries, years with a higher lagged internet-user share were not years of faster productivity growth. The linear estimate is negative and stable across trends, convergence controls and development-specific year effects; the flexible estimate is negative in almost every specification but its interval includes zero with two-way clustering, for some split seeds and in many robustness checks. We therefore do not claim a significant negative association. What the data rule out, under our assumptions, is a positive within-country association of the size reported by earlier macro studies. The association is concentrated in 1996–2011, is weaker without the transition economies, and is shared with the co-diffusing mobile and broadband series. It is a low-frequency co-movement between the level of adoption and growth, and changes in adoption are not associated with growth. The literature suggests several explanations, which our design cannot distinguish: the J-curve argument that intangible investment and reorganisation depress measured productivity during diffusion (Brynjolfsson et al., 2021), the concentration of ICT returns in firms with complementary capabilities (Gal et al., 2019; Nicoletti et al., 2020), confounding by the growth slowdown that followed the 2000s boom in economies that adopted late, and other unobserved confounding. A shift of employment towards low-productivity activities does not fit the decomposition, because the association is with output growth rather than with employment growth. A non-positive within-country association is also consistent with a positive cross-country association in long differences, which reflects which economies adopted, not what adoption did within them.

Second, none of the heterogeneity predictions is supported robustly. With our precision, we cannot distinguish a flat function from moderate gradients of either sign, so the absence of evidence for complementarities with schooling is not evidence of their absence. Along initial productivity, some flexible fits suggest non-monotone variation, but the evidence is weak and not of the form the theories predict. The J-curve pattern along time since take-off

that appears in the complete-ICT sample does not survive in the full sample.

Third, estimates of where the association differs are sensitive to the data. The gradient along initial productivity is large in both PWT releases and much weaker in WDI data for the same country-years. On identical country-years, about half of the difference is due to the growth series, whose within-country agreement between the two sources is weakest in the least productive economies, and a third to the control set; the difference is stable across split seeds but imprecise. The revision between PWT releases, by contrast, does not change the gradient detectably. These findings extend the evidence of Johnson et al. (2013) and Ponomareva and Katayama (2010) on data choices to heterogeneous associations of internet diffusion, and they are a warning against reading such gradients as stable facts. The linear benchmarks point the other way in the WDI data: TWFE with a linear interaction gives a tercile contrast of 1.77 (0.85) along initial productivity and TWFE with spline controls 3.90 (1.14). In the plasmode design without any effect, the same estimators produce gradients of similar size and reject in most replications (Table C.9), so we do not take this as evidence of a gradient.

6.2. Policy implications

Because the estimates are conditional associations, their policy content is limited, and they do not identify the return to connectivity investment. With that qualification, they caution against justifying connectivity programmes by expected short-run gains in aggregate measured productivity: in the economies and period we study, years with higher internet use were not followed by faster measured productivity growth within countries over horizons of up to nine years. This does not imply that such programmes are undesirable. Connectivity has consumer-surplus, inclusion and service-delivery benefits that productivity statistics do not capture (World Bank, 2016; Goldfarb and Tucker, 2019; Hjort and Tian, 2025), and averages across countries can hide large gains for specific groups within them. The literature offers several explanations for a missing aggregate pay-off that our data cannot test: gains may depend on complementary investment in firm capabilities, management and skills

(Bloom et al., 2012; Gal et al., 2019), on the intensity and quality of use rather than on access (Comin and Mestieri, 2018), or on reorganisation that takes longer than our horizons (Brynjolfsson et al., 2021). These are hypotheses for further work, not implications of our estimates. Finally, cross-country benchmarks that claim to identify the stage of development at which digital investment pays off should be treated with caution, because such heterogeneity estimates change with data releases. Policy-relevant causal statements would require designs with credible exogenous variation, such as the staggered arrival of infrastructure used by Hjort and Poulsen (2019).

6.3. Limitations

The estimates are conditional associations of a lagged adoption level with growth, and the assumptions behind them are strong. Strict exogeneity rules out feedback from productivity shocks to future adoption; the lead test does not reject it but has little power, lagging the treatment by one or two years mitigates but does not eliminate reverse causality, and the instrument we could construct from historical fixed-line density is weak. Nuisance sufficiency requires that the controls and their country means capture the confounding path, which rules out, for example, global shocks with country-specific loadings that are correlated with adoption, or development trends that the controls do not track; region-by-year effects address the first concern only partly. The baseline avoids the mechanical violation of strict exogeneity that lagged productivity controls create, but other controls, such as urbanisation, may respond to past productivity, and convergence dynamics are absorbed only to the extent that initial productivity and the country means capture them. Measurement error matters in both directions: employment in low-income economies is largely modelled by the ILO and ITU internet shares for these economies are often estimated. In a linear model classical error in adoption would attenuate the coefficient; in our residualised varying-coefficient model the direction of the bias is not known in general. Results are similar with GDP-per-capita growth as the outcome, in countries with above-median statistical capacity, and without screening the adoption series, but they cannot rule out non-classical errors correlated with

development. The last observations rest on provisional national accounts and schooling in 2024 is extrapolated. Internet penetration measures access, not intensity or quality of use; for low-income economies, mobile-network coverage and the quality of connections (3G or 4G) may matter more than the share of users, and ITU user shares for many of them are estimated from subscriptions rather than measured in household surveys. For economies without labour-force surveys, the ILO models employment with predictors that include demographic and urbanisation variables, two of which are among our controls; errors in the denominator of productivity can therefore be correlated with the controls. Productivity statistics also miss the value of free digital goods, and aggregate data cannot reveal effects that differ by gender, between rural and urban areas, or between formal and informal firms. Coverage of low-income and conflict-affected economies is incomplete (14 of 25 low-income economies are included), so the policy caution applies mainly to middle- and high-income economies. Finally, the asymptotic theory covers the fixed- K sieve with a vanishing penalty and, for the shape tests, a fixed penalty (Propositions 3 and 4); the data-dependent penalty is evaluated by simulation only, and the bands describe the penalised fit rather than θ itself (Remark 1).

7. Conclusion

Over three decades of internet diffusion in 149 economies, we do not find a positive within-country association between lagged internet use and labour-productivity growth. The linear estimate is negative and robust to trends and convergence controls; the flexible estimate is negative but not clearly different from zero, and the association is concentrated in the first half of the period. We find no robust variation of the association with schooling, initial productivity or time since take-off, within the precision of the data. Heterogeneity along the development dimension appears in the Penn World Table and is much weaker in World Bank data for the same country-years, largely because the growth series differ, while the revision between PWT releases does not change it detectably. These are conditional associations,

not causal effects of connectivity.

Methodologically, Panel-DOSE combines orthogonal series estimation, Mundlak features, country-level cross-fitting, a within transformation and an O’Sullivan penalty, whose null space makes the constancy and linearity tests valid for a fixed penalty. In a plasmode design built on our data, the constancy test with the cross-validated penalty has size between 0.06 and 0.09 at the 5% level; finite-sample calibration of this kind should accompany applications, because the theory does not cover the data-dependent penalty. In our simulations it is most useful when confounding is strongly nonlinear; when it is mild, TWFE with spline controls does as well. Most of its advantage comes from the Mundlak features, and linear interaction models give unreliable intervals when the control function is misspecified. The code, which also reports identifying variation, paired-bootstrap comparisons of estimates and shape tests, is provided as an installable Python module in the replication package.

Future work could combine the estimator with instruments or staggered infrastructure roll-outs to move from associations to effects, use firm- or region-level measures of the intensity of digital use, and examine the moderating roles of institutions, competition and data governance.

Data and code availability

The data are publicly available from the World Bank’s World Development Indicators (July 2026 release), the UNDP Human Development Report 2025 and the Penn World Table (versions 11.0 and 10.0). The WDI snapshot used here was retrieved through the World Bank API in September 2026 and is archived with SHA-256 checksums in the replication package, which reads the archived snapshot by default and verifies it before the panels are built. The package contains the Python implementation of Panel-DOSE, the scripts that reproduce all tables, figures and numbers in the text, and the list of package versions used. The package is available to the editor and reviewers; its location is given on the title page to preserve anonymity.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author(s) used generative AI tools to assist with writing and debugging code and with drafting and editing the text. The research idea, the model and the algorithm are the author(s)' own. After using these tools, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

References

- Acemoglu, D., Aghion, P., Zilibotti, F., 2006. Distance to frontier, selection, and economic growth. *Journal of the European Economic Association* 4, 37–74. doi:10.1162/jeea.2006.4.1.37.
- Akerman, A., Gaarder, I., Mogstad, M., 2015. The skill complementarity of broadband internet. *The Quarterly Journal of Economics* 130, 1781–1824. doi:10.1093/qje/qjv028.
- Angrist, N., Djankov, S., Goldberg, P.K., Patrinos, H.A., 2021. Measuring human capital using global learning data. *Nature* 592, 403–408. doi:10.1038/s41586-021-03323-7.
- Athey, S., Tibshirani, J., Wager, S., 2019. Generalized random forests. *The Annals of Statistics* 47, 1148–1178. doi:10.1214/18-aos1709.
- Banga, K., Harbansh, P., Singh, S., 2026. Digitalisation and structural change: Evidence from cross-country analysis. *Journal of Digital Economy* 5, 314–329. doi:10.1016/j.jdec.2026.03.003.
- Belloni, A., Chernozhukov, V., Chetverikov, D., Kato, K., 2015. Some new asymptotic theory for least squares series: Pointwise and uniform results. *Journal of Econometrics* 186, 345–366. doi:10.1016/j.jeconom.2015.02.014.
- Benhabib, J., Spiegel, M.M., 1994. The role of human capital in economic development: Evidence from aggregate cross-country data. *Journal of Monetary Economics* 34, 143–173. doi:10.1016/0304-3932(94)90047-7.
- Bloom, N., Sadun, R., Van Reenen, J., 2012. Americans do IT better: US multinationals and the productivity miracle. *American Economic Review* 102, 167–201. doi:10.1257/aer.102.1.167.
- Bresnahan, T.F., Brynjolfsson, E., Hitt, L.M., 2002. Information technology, workplace organization, and the demand for skilled labor: Firm-level evidence. *The Quarterly Journal of Economics* 117, 339–376. doi:10.1162/003355302753399526.

- Brynjolfsson, E., Hitt, L.M., 2000. Beyond computation: Information technology, organizational transformation and business performance. *Journal of Economic Perspectives* 14, 23–48. doi:10.1257/jep.14.4.23.
- Brynjolfsson, E., Rock, D., Syverson, C., 2021. The productivity J-curve: How intangibles complement general purpose technologies. *American Economic Journal: Macroeconomics* 13, 333–372. doi:10.1257/mac.20180386.
- Byrne, D.M., Fernald, J.G., Reinsdorf, M.B., 2016. Does the United States have a productivity slowdown or a measurement problem? *Brookings Papers on Economic Activity* 2016, 109–182. doi:10.1353/eca.2016.0014.
- Callaway, B., Goodman-Bacon, A., Sant’Anna, P.H.C., 2024. Difference-in-differences with a continuous treatment. NBER Working Paper 32117. National Bureau of Economic Research. doi:10.3386/w32117.
- Cameron, A.C., Gelbach, J.B., Miller, D.L., 2011. Robust inference with multiway clustering. *Journal of Business & Economic Statistics* 29, 238–249. doi:10.1198/jbes.2010.07136.
- Cameron, A.C., Miller, D.L., 2015. A practitioner’s guide to cluster-robust inference. *Journal of Human Resources* 50, 317–372. doi:10.3368/jhr.50.2.317.
- Cardona, M., Kretschmer, T., Stobbel, T., 2013. ICT and productivity: Conclusions from the empirical literature. *Information Economics and Policy* 25, 109–125. doi:10.1016/j.infoecopol.2012.12.002.
- de Chaisemartin, C., D’Haultfœuille, X., 2020. Two-way fixed effects estimators with heterogeneous treatment effects. *American Economic Review* 110, 2964–2996. doi:10.1257/aer.20181169.
- Chen, X., Li, Y., 2026. Aggregate production functions in the digital economy: A survey of microfoundational approaches. *Journal of Digital Economy* 5, 87–98. doi:10.1016/j.jdec.2025.12.003.
- Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., Robins, J., 2018. Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal* 21, C1–C68. doi:10.1111/ectj.12097.
- Clarke, P.S., Polsell, A., 2026. Double machine learning for static panel models with fixed effects. *The Econometrics Journal* 29, 69–86. doi:10.1093/ectj/utaf011.
- Cohen, W.M., Levinthal, D.A., 1990. Absorptive capacity: A new perspective on learning and innovation. *Administrative Science Quarterly* 35, 128–152. doi:10.2307/2393553.
- Colangelo, K., Lee, Y.Y., 2026. Double debiased machine learning nonparametric inference with continuous treatments. *Journal of Business & Economic Statistics* 44, 67–79. doi:10.1080/07350015.2025.2505487.

- Comin, D., Hobijn, B., 2010. An exploration of technology diffusion. *American Economic Review* 100, 2031–2059. doi:10.1257/aer.100.5.2031.
- Comin, D., Mestieri, M., 2018. If technology has arrived everywhere, why has income diverged? *American Economic Journal: Macroeconomics* 10, 137–178. doi:10.1257/mac.20150175.
- Czernich, N., Falck, O., Kretschmer, T., Woessmann, L., 2011. Broadband infrastructure and economic growth. *The Economic Journal* 121, 505–532. doi:10.1111/j.1468-0297.2011.02420.x.
- Eilers, P.H.C., Marx, B.D., 1996. Flexible smoothing with B-splines and penalties. *Statistical Science* 11, 89–121. doi:10.1214/ss/1038425655.
- Feenstra, R.C., Inklaar, R., Timmer, M.P., 2015. The next generation of the Penn World Table. *American Economic Review* 105, 3150–3182. doi:10.1257/aer.20130954.
- Franklin, J.M., Schneeweiss, S., Polinski, J.M., Rassen, J.A., 2014. Plasmode simulation for the evaluation of pharmacoepidemiologic methods in complex healthcare databases. *Computational Statistics & Data Analysis* 72, 219–226. doi:10.1016/j.csda.2013.10.018.
- Gal, P., Nicoletti, G., Renault, T., Sorbe, S., Timiliotis, C., 2019. Digitalisation and productivity: In search of the holy grail – Firm-level empirical evidence from EU countries. OECD Economics Department Working Papers 1533. OECD Publishing.
- Goldfarb, A., Tucker, C., 2019. Digital economics. *Journal of Economic Literature* 57, 3–43.
- Groningen Growth and Development Centre, 2025. Penn World Table version 11.0. Data-verseNL. doi:10.34894/FABVLR. University of Groningen, <https://www.rug.nl/ggdc/productivity/pwt/>.
- Hjort, J., Poulsen, J., 2019. The arrival of fast internet and employment in Africa. *American Economic Review* 109, 1032–1079.
- Hjort, J., Tian, L., 2025. The economic impact of internet connectivity in developing countries. *Annual Review of Economics* 17, 99–124. doi:10.1146/annurev-economics-081224-102352.
- Holm, S., 1979. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics* 6, 65–70.
- Islam, N., 1995. Growth empirics: A panel data approach. *The Quarterly Journal of Economics* 110, 1127–1170. doi:10.2307/2946651.
- Johnson, S., Larson, W., Papageorgiou, C., Subramanian, A., 2013. Is newer better? Penn World Table revisions and their impact on growth estimates. *Journal of Monetary Economics* 60, 255–274. doi:10.1016/j.jmoneco.2012.10.022.

- Jordà, Ò., 2005. Estimation and inference of impulse responses by local projections. *American Economic Review* 95, 161–182. doi:10.1257/0002828053828518.
- Jorgenson, D.W., Ho, M.S., Stiroh, K.J., 2008. A retrospective look at the U.S. productivity growth resurgence. *Journal of Economic Perspectives* 22, 3–24.
- Jorgenson, D.W., Vu, K.M., 2016. The ICT revolution, world economic growth, and policy issues. *Telecommunications Policy* 40, 383–397. doi:10.1016/j.telpol.2016.01.002.
- Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., Liu, T.Y., 2017. LightGBM: A highly efficient gradient boosting decision tree, in: *Advances in Neural Information Processing Systems*, pp. 3146–3154.
- Klosin, S., Vilgalys, M., 2022. Estimating continuous treatment effects in panel data using machine learning with a climate application. *arXiv preprint arXiv:2207.08789*.
- Koutroumpis, P., 2009. The economic impact of broadband on growth: A simultaneous approach. *Telecommunications Policy* 33, 471–485. doi:10.1016/j.telpol.2009.07.004.
- Montiel Olea, J.L., Plagborg-Møller, M., 2019. Simultaneous confidence bands: Theory, implementation, and an application to SVARs. *Journal of Applied Econometrics* 34, 1–17. doi:10.1002/jae.2656.
- Mundlak, Y., 1978. On the pooling of time series and cross section data. *Econometrica* 46, 69–85.
- Nelson, R.R., Phelps, E.S., 1966. Investment in humans, technological diffusion, and economic growth. *American Economic Review* 56, 69–75.
- Nicoletti, G., von Rueden, C., Andrews, D., 2020. Digital technology diffusion: A matter of capabilities, incentives or both? *European Economic Review* 128, 103513. doi:10.1016/j.euroecorev.2020.103513.
- Nie, X., Wager, S., 2021. Quasi-oracle estimation of heterogeneous treatment effects. *Biometrika* 108, 299–319. doi:10.1093/biomet/asaa076.
- Niebel, T., 2018. ICT and economic growth – comparing developing, emerging and developed countries. *World Development* 104, 197–211. doi:10.1016/j.worlddev.2017.11.024.
- O’Sullivan, F., 1986. A statistical perspective on ill-posed inverse problems. *Statistical Science* 1, 502–518. doi:10.1214/ss/1177013525.
- Ponomareva, N., Katayama, H., 2010. Does the version of the Penn World Tables matter? An analysis of the relationship between growth and volatility. *Canadian Journal of Economics* 43, 152–179. doi:10.1111/j.1540-5982.2009.01567.x.
- Robinson, P.M., 1988. Root-N-consistent semiparametric regression. *Econometrica* 56, 931–954. doi:10.2307/1912705.

- Röller, L.H., Waverman, L., 2001. Telecommunications infrastructure and economic development: A simultaneous approach. *American Economic Review* 91, 909–923. doi:10.1257/aer.91.4.909.
- Semenova, V., Chernozhukov, V., 2021. Debiased machine learning of conditional average treatment effects and other causal functions. *The Econometrics Journal* 24, 264–289. doi:10.1093/ectj/utaa027.
- Semenova, V., Goldman, M., Chernozhukov, V., Taddy, M., 2023. Inference on heterogeneous treatment effects in high-dimensional dynamic panels under weak dependence. *Quantitative Economics* 14, 471–510. doi:10.3982/QE1670.
- Syverson, C., 2017. Challenges to mismeasurement explanations for the US productivity slowdown. *Journal of Economic Perspectives* 31, 165–186. doi:10.1257/jep.31.2.165.
- Vu, K., Hanafizadeh, P., Bohlin, E., 2020. ICT as a driver of economic growth: A survey of the literature and directions for future research. *Telecommunications Policy* 44, 101922. doi:10.1016/j.telpol.2020.101922.
- Wager, S., Athey, S., 2018. Estimation and inference of heterogeneous treatment effects using random forests. *Journal of the American Statistical Association* 113, 1228–1242. doi:10.1080/01621459.2017.1319839.
- Wand, M.P., Ormerod, J.T., 2008. On semiparametric regression with O’Sullivan penalized splines. *Australian & New Zealand Journal of Statistics* 50, 179–198.
- Wooldridge, J.M., 2010. *Econometric Analysis of Cross Section and Panel Data*. 2nd ed., MIT Press, Cambridge, MA.
- World Bank, 2016. *World Development Report 2016: Digital Dividends*. World Bank, Washington, DC.
- Xu, Y., Wei, P., Ji, Y., 2026. Machine learning for predicting corporate carbon emissions: The role of corporate governance. *Journal of Digital Economy* 5, 1–16. doi:10.1016/j.jdec.2025.11.008.